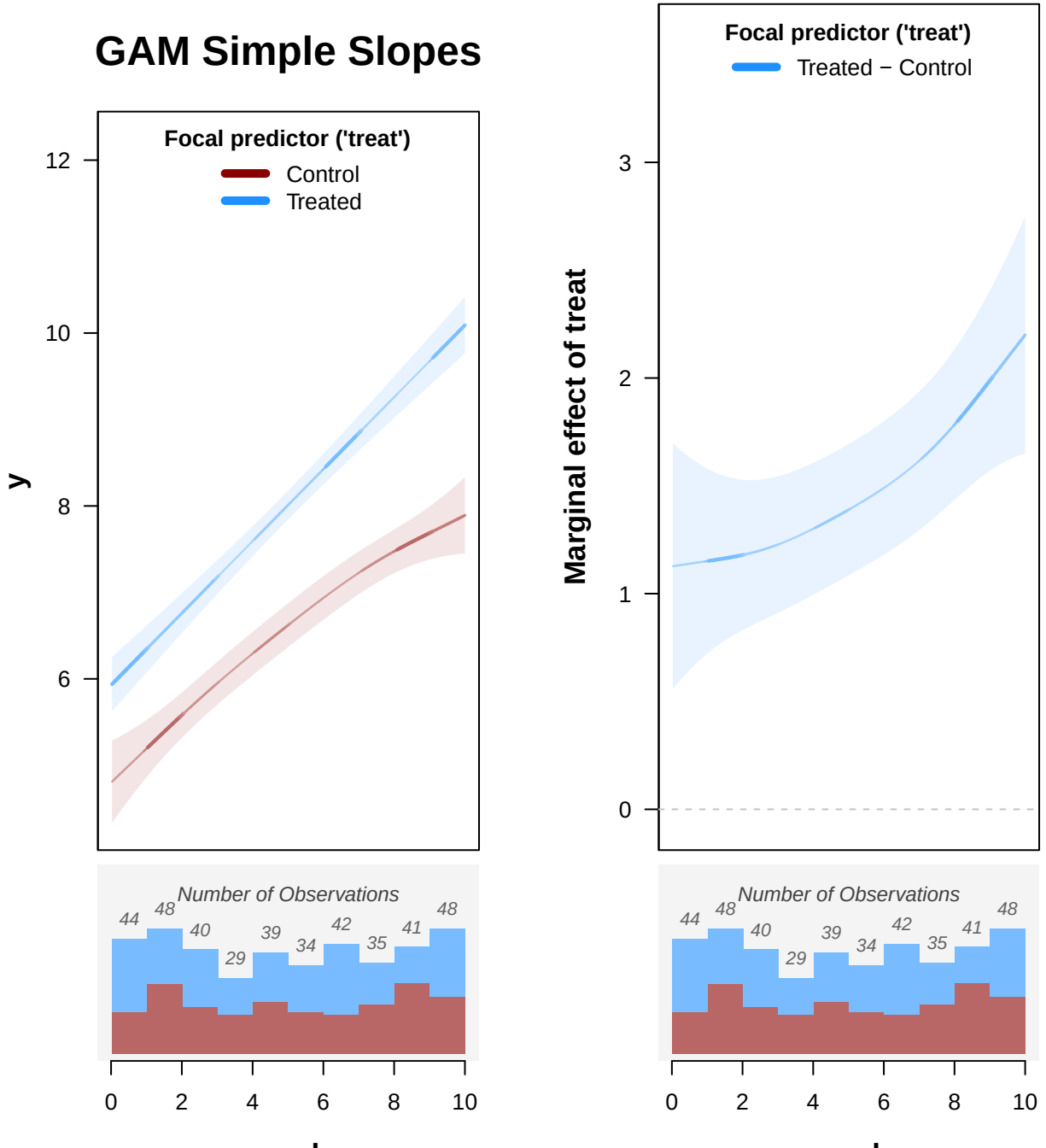
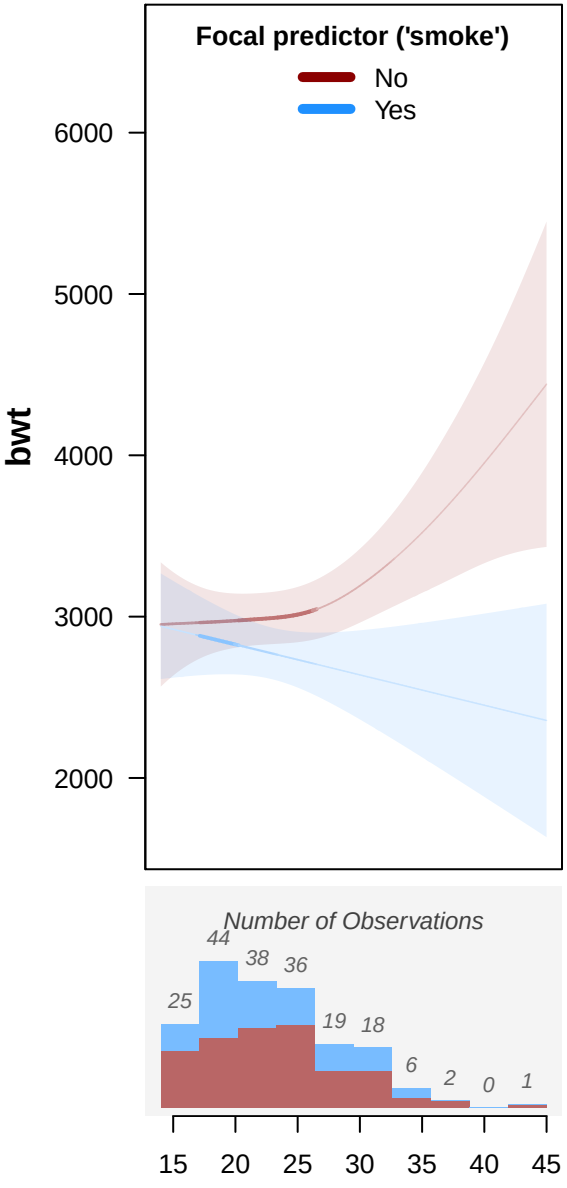
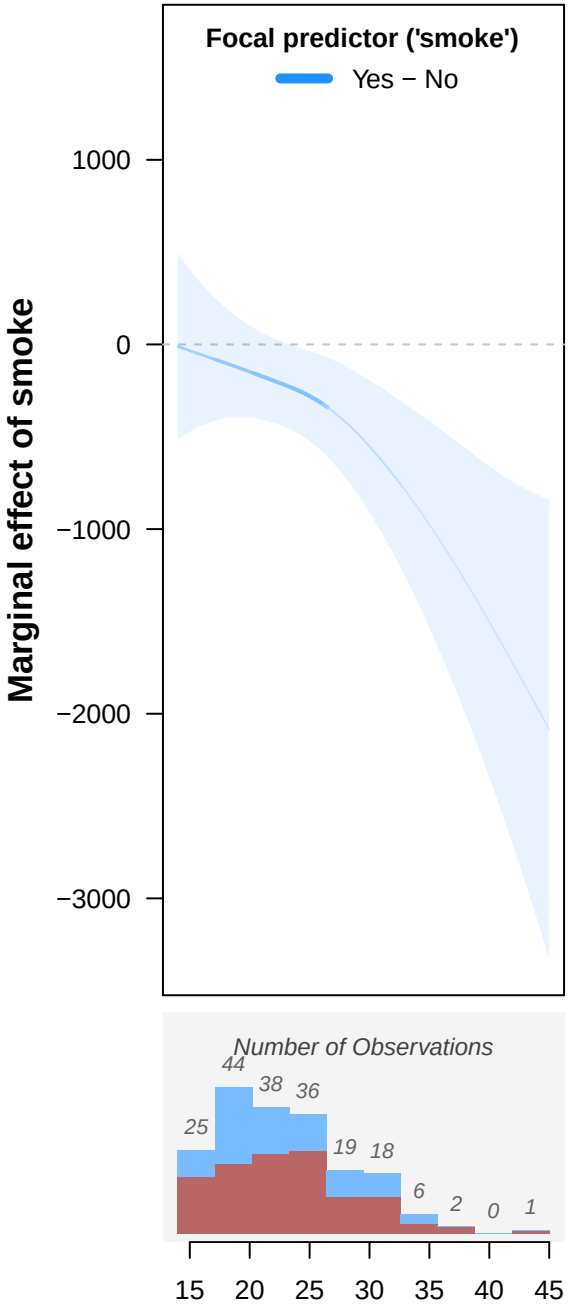

GAM Simple Slopes
Focal predictor ('treat')
Control
Treated
y
12
10
8
6
Number of Observations
44
48
40
29
39
34
42
35
41
48
0
2
4
6
8
10
Focal predictor ('treat')
Treated − Control
Marginal effect of treat
3
2
1
0
Number of Observations
44
48
40
29
39
34
42
35
41
48
0
2
4
6
8
10
help("interprobe")

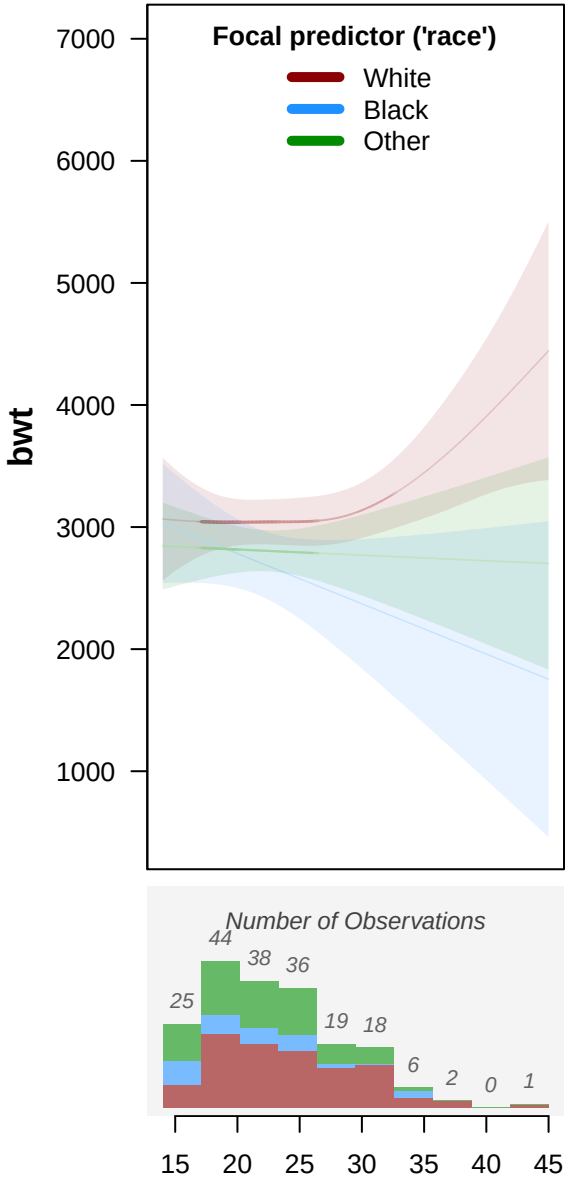
GAM Simple Slopes

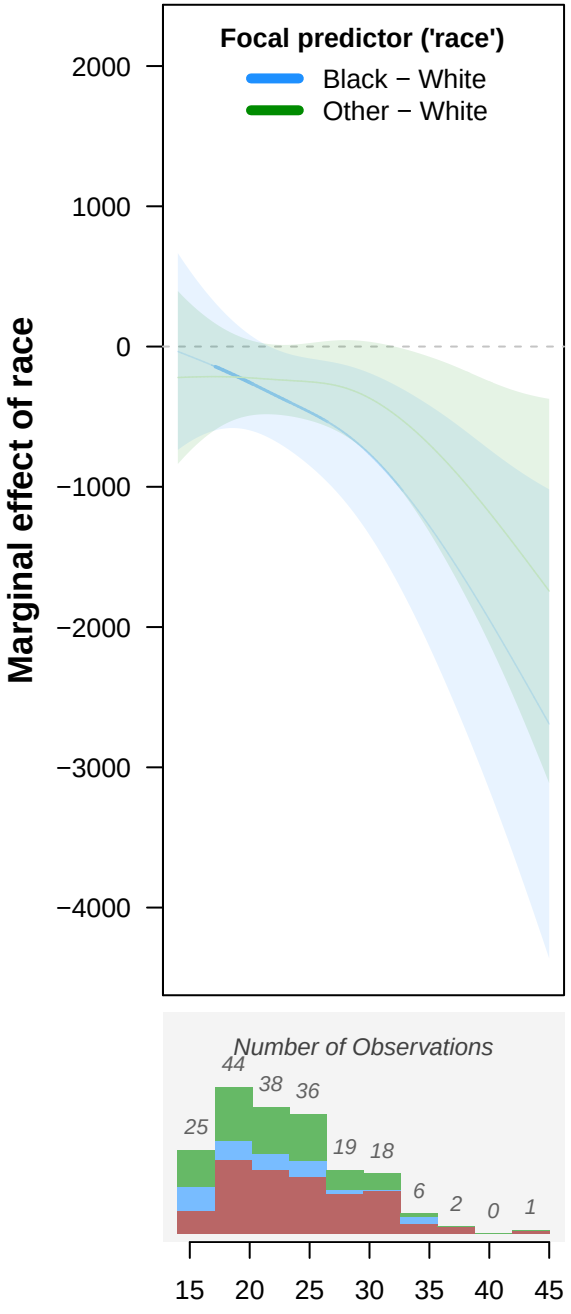




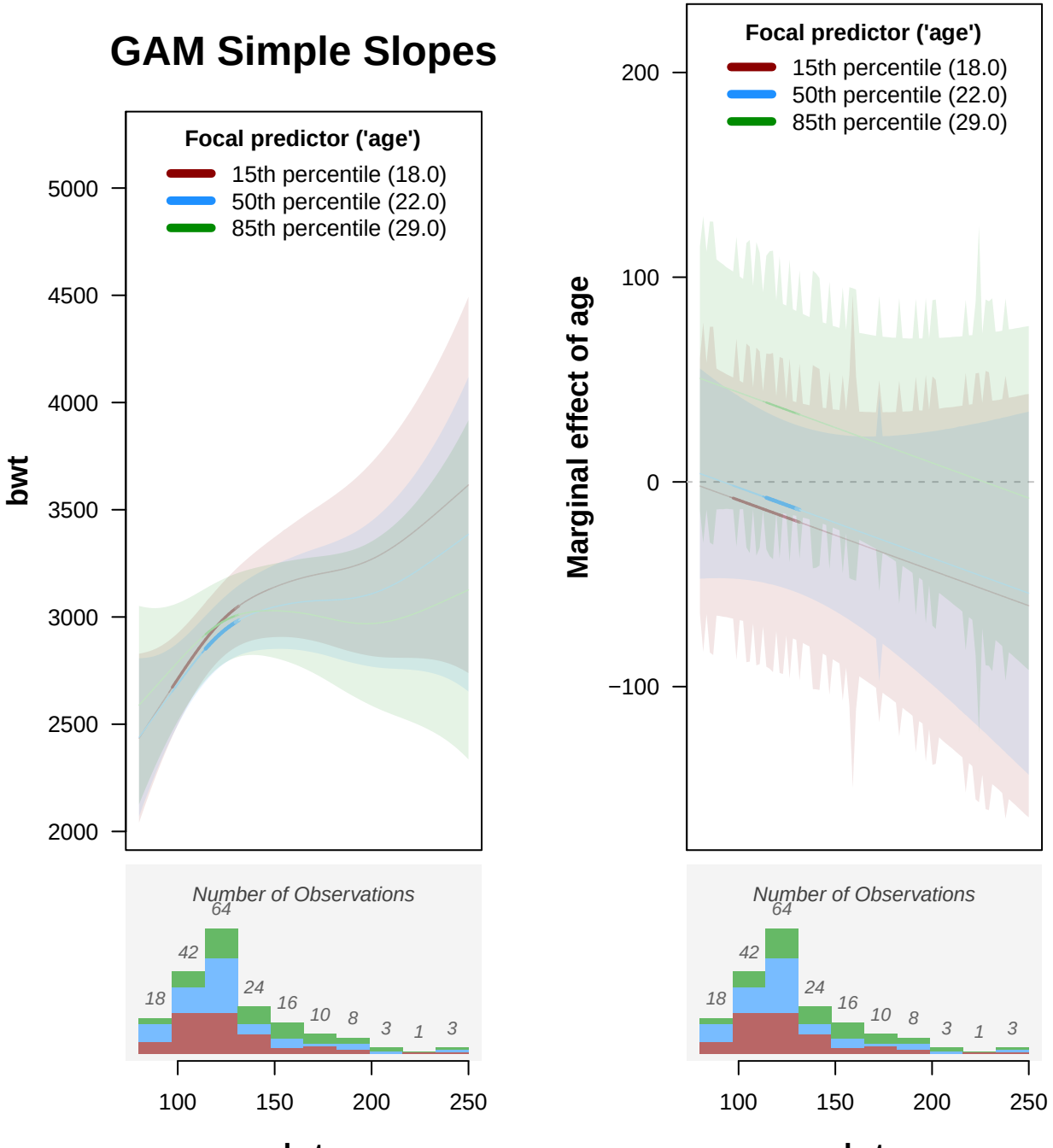
help("interprobe")

GAM Simple Slopes

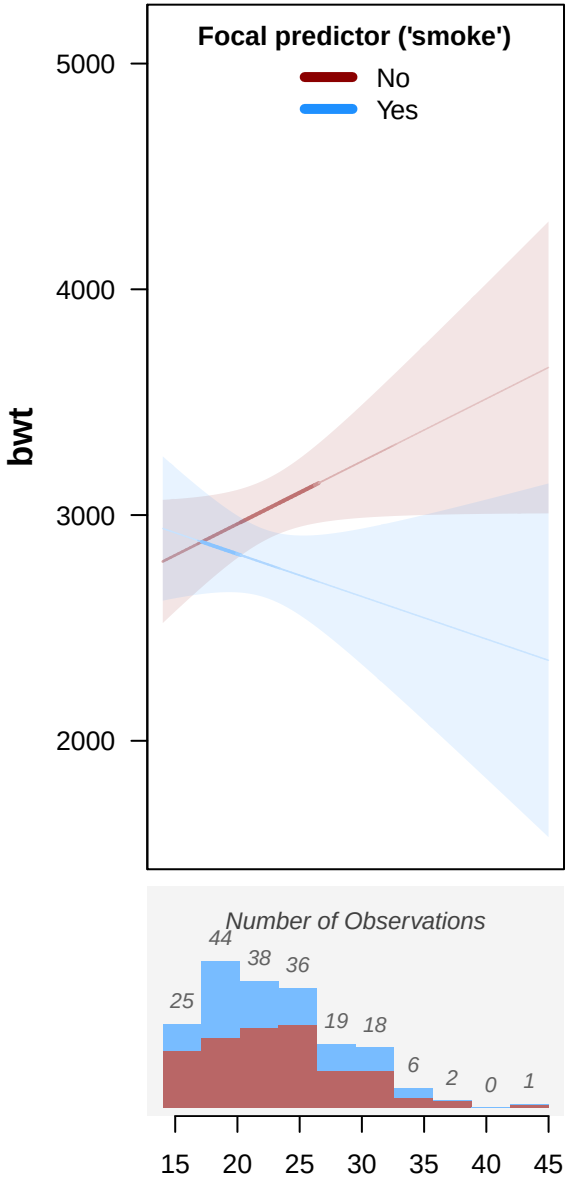


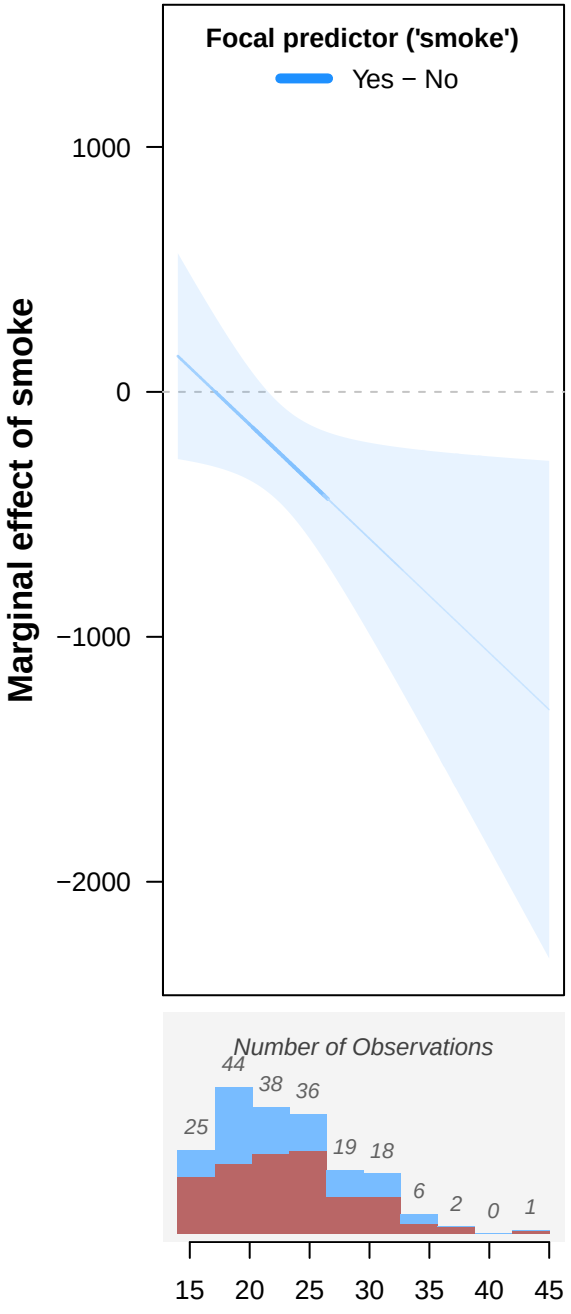


help("interprobe")


GAM Simple Slopes
Focal predictor ('age')
15th percentile (18.0)
50th percentile (22.0)
85th percentile (29.0)
bwt
5000
4500
4000
3500
3000
2500
2000
Number of Observations
64
42
18
24
16
10
8
3
1
3
100
150
200
250
Marginal effect of age
200
100
0
−100
help("interprobe")

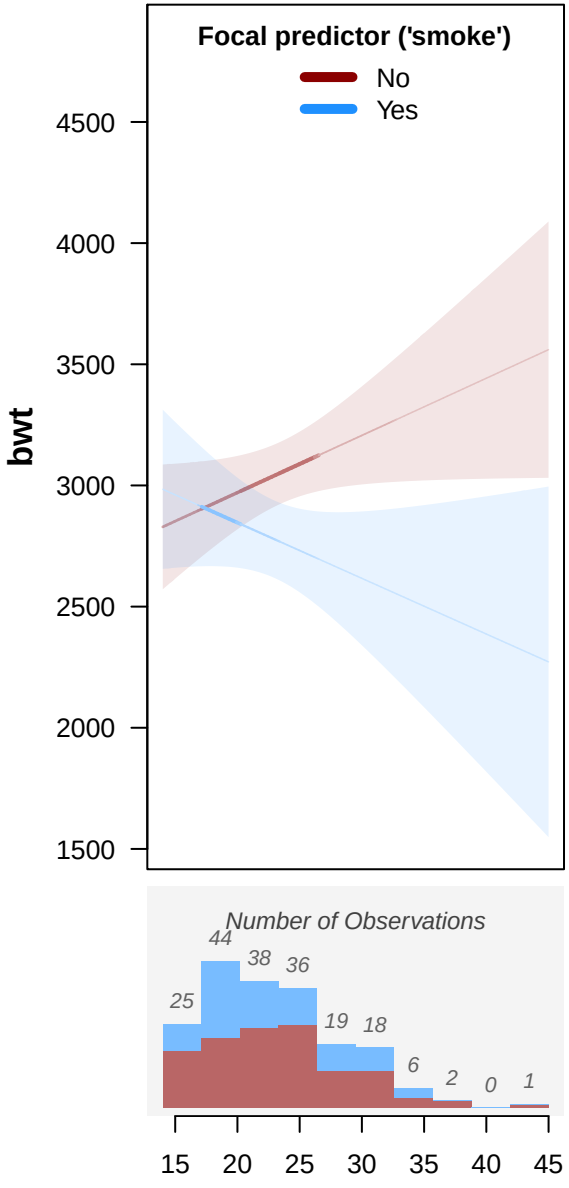
GAM Simple Slopes

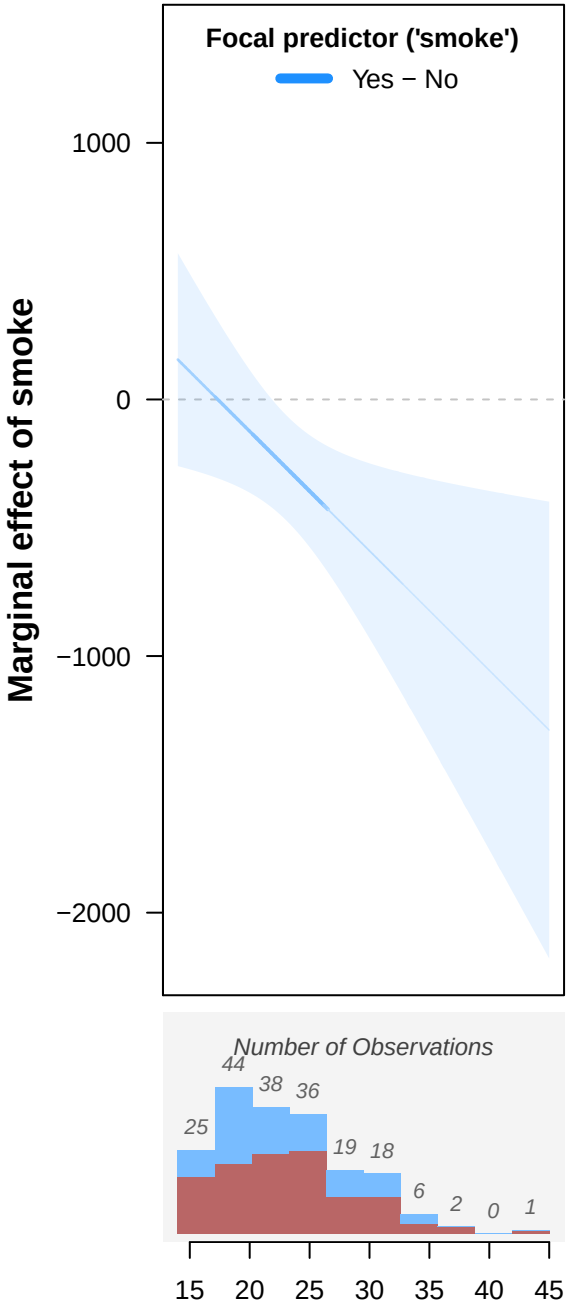




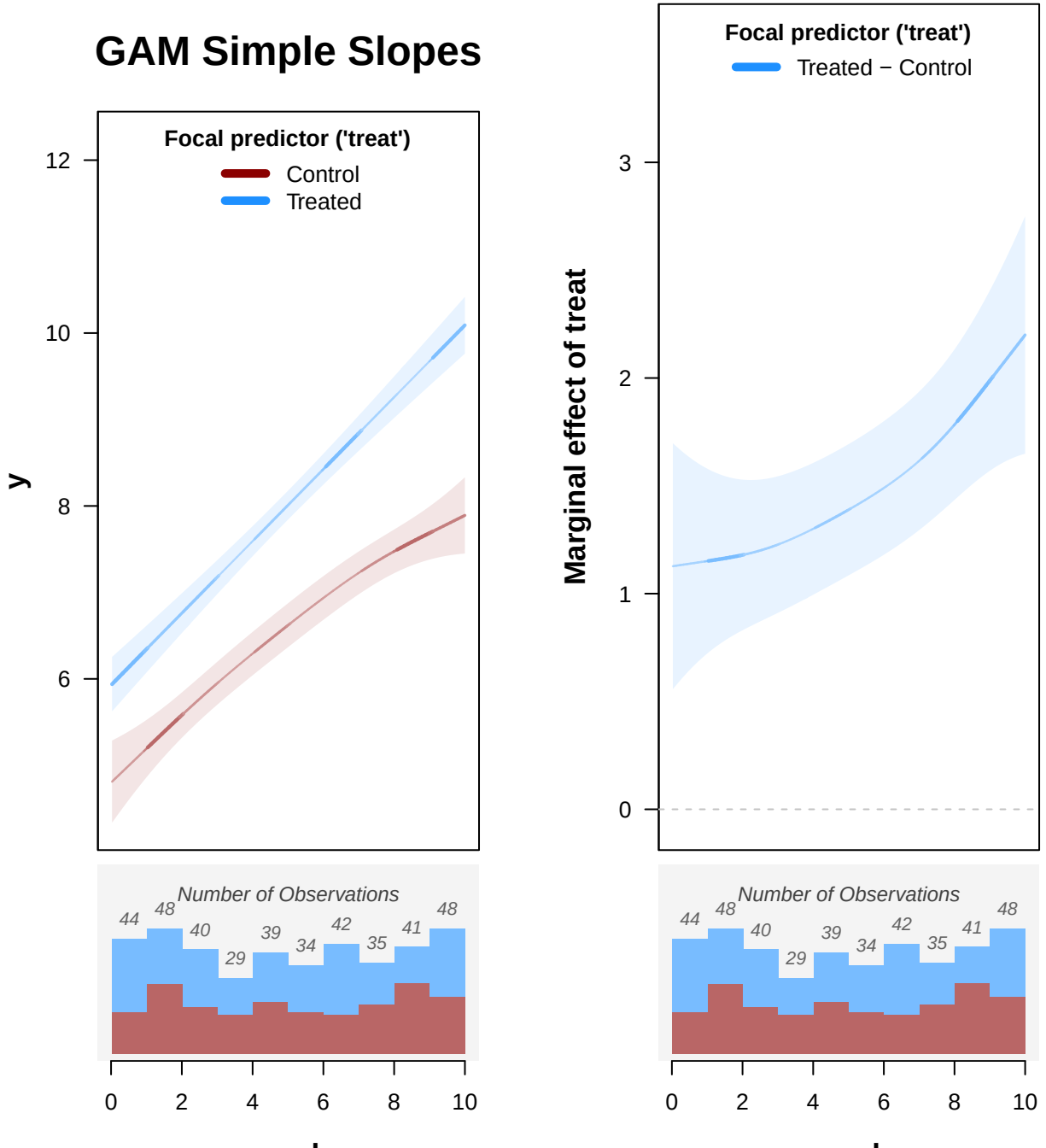
help("interprobe")

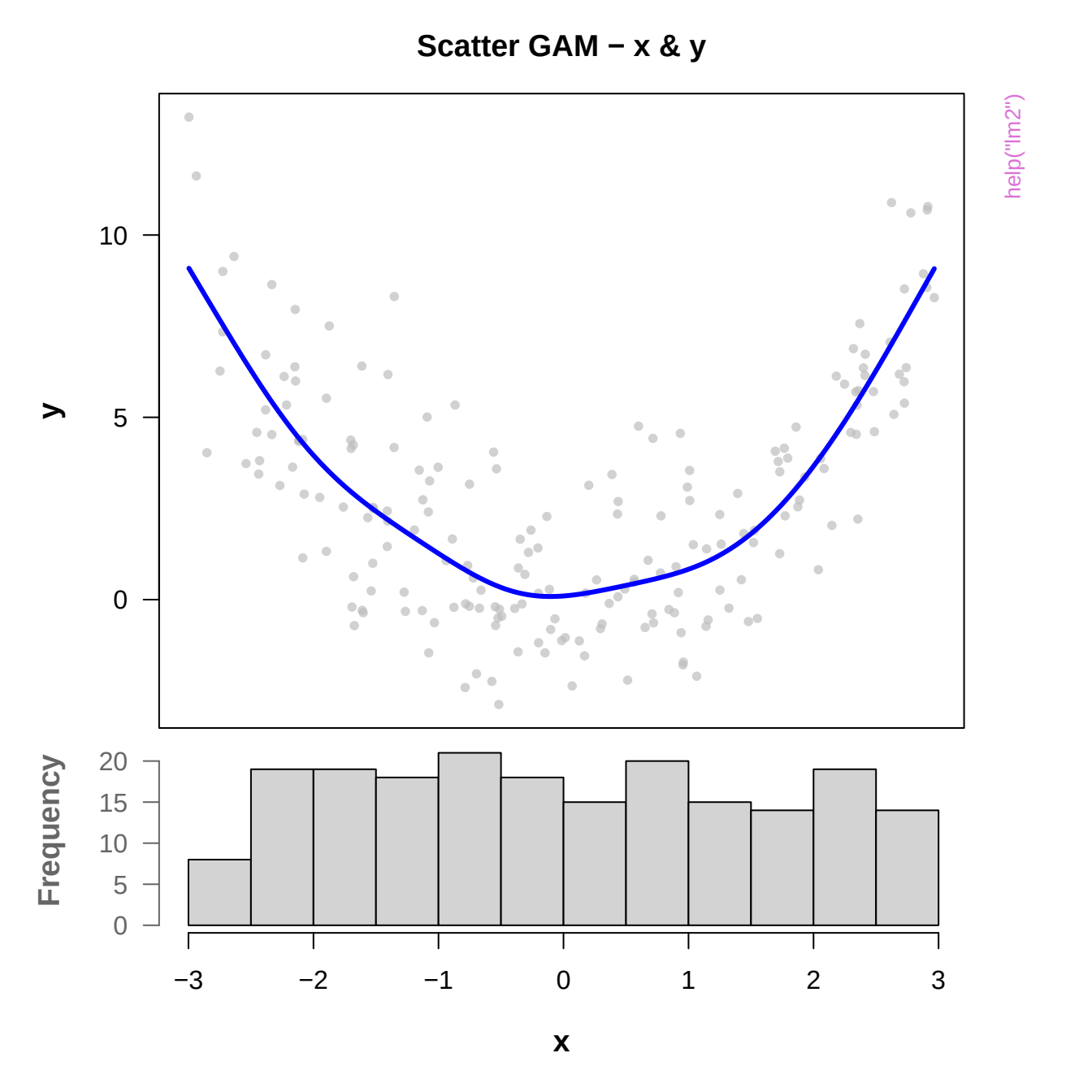
Linear Simple Slopes

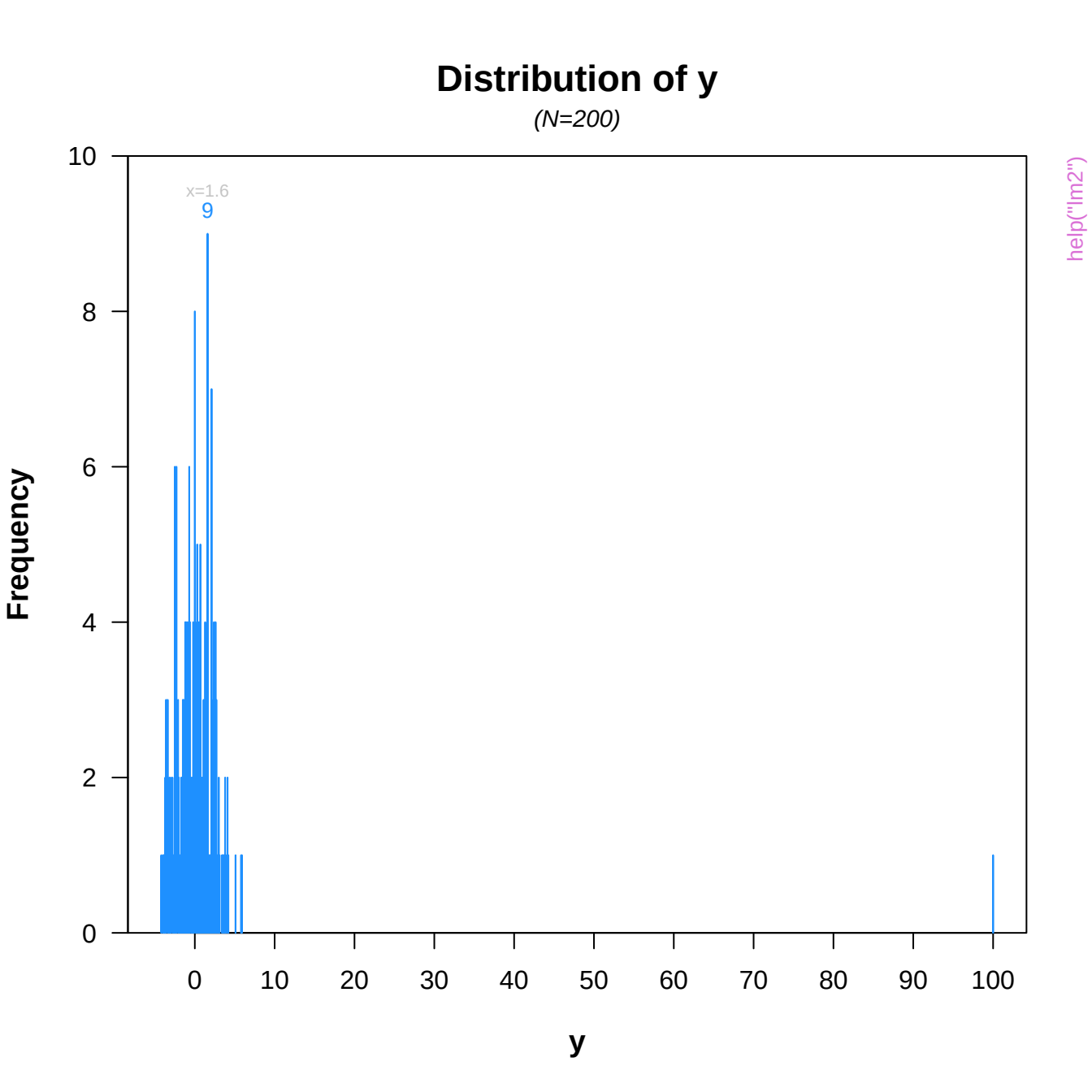


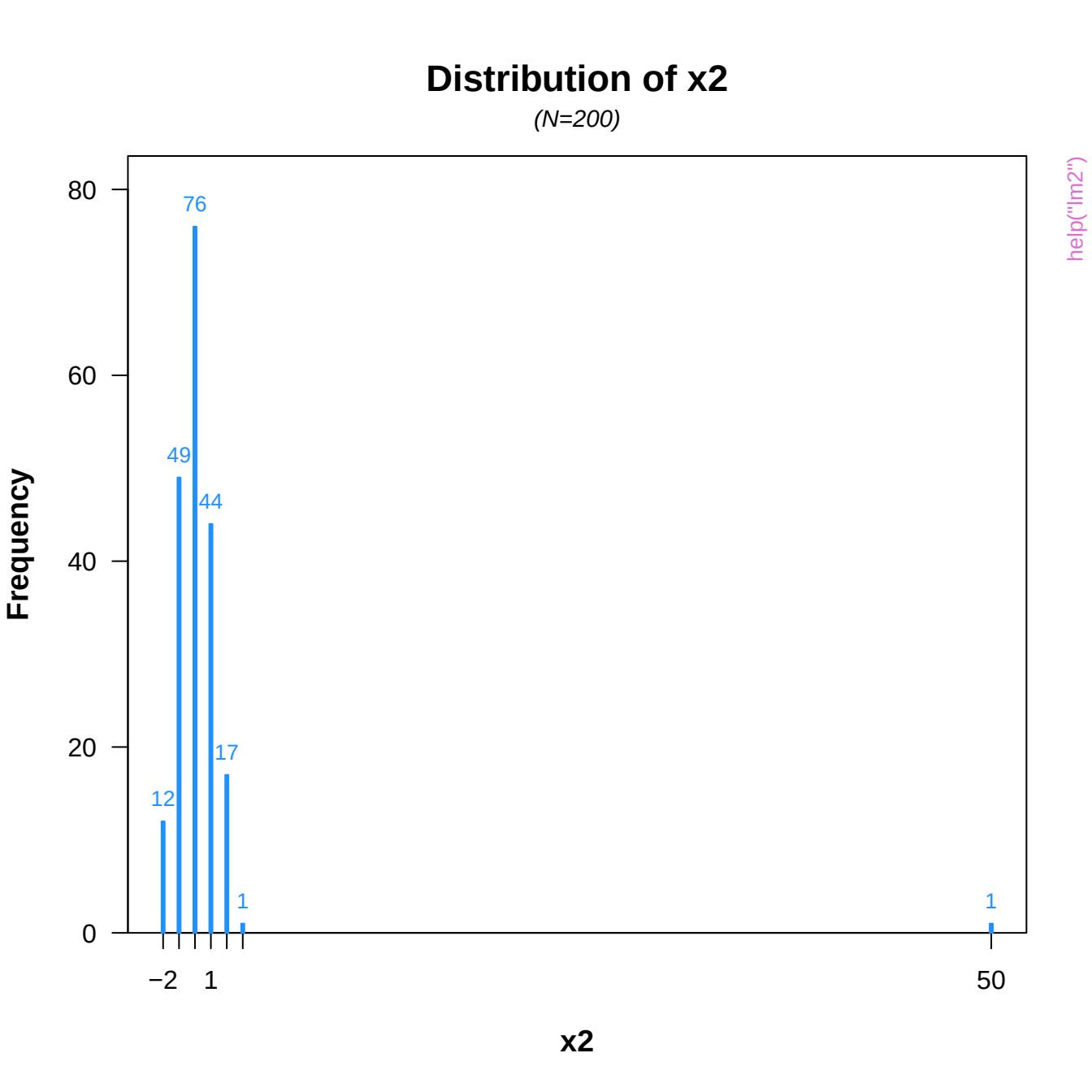


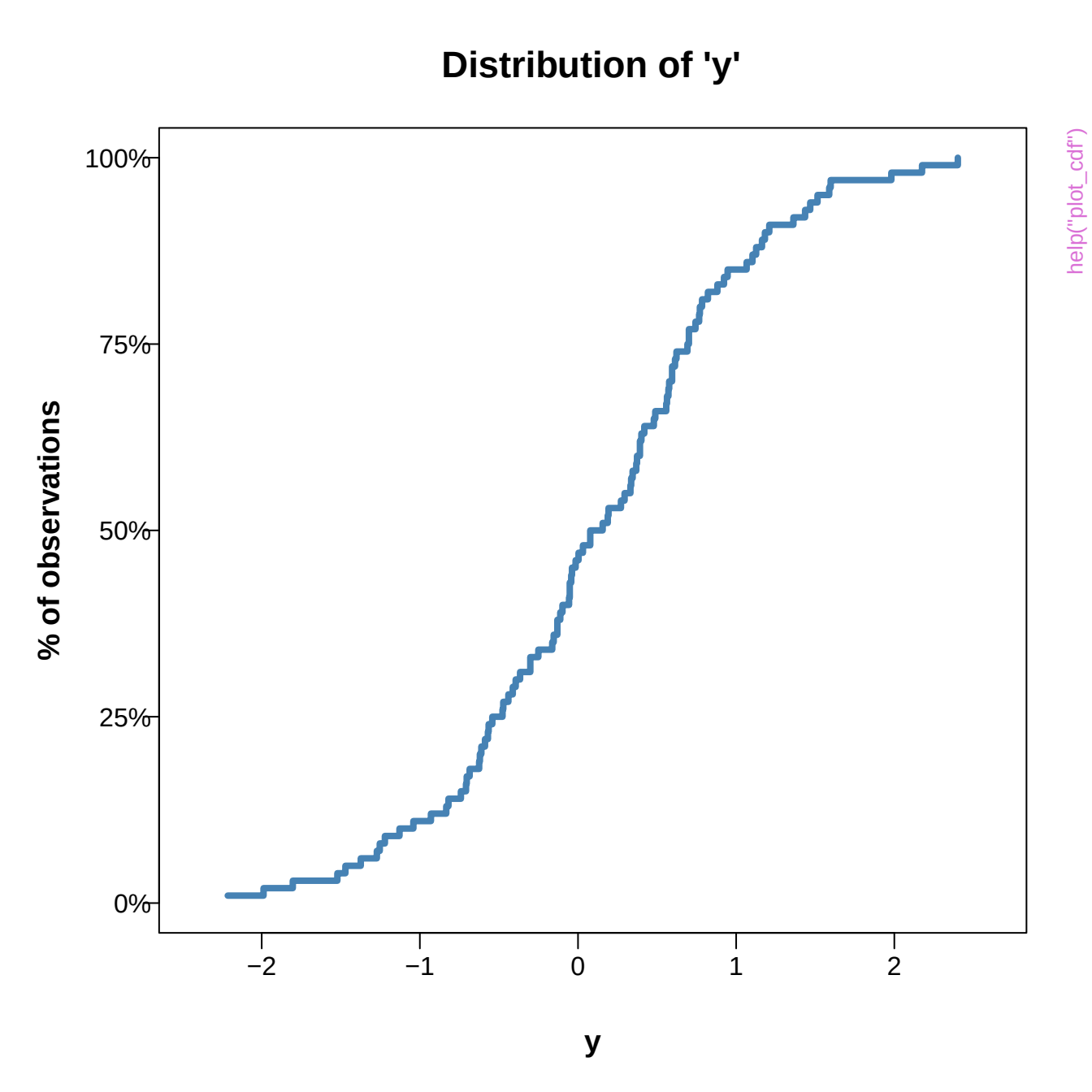
help("interprobe")

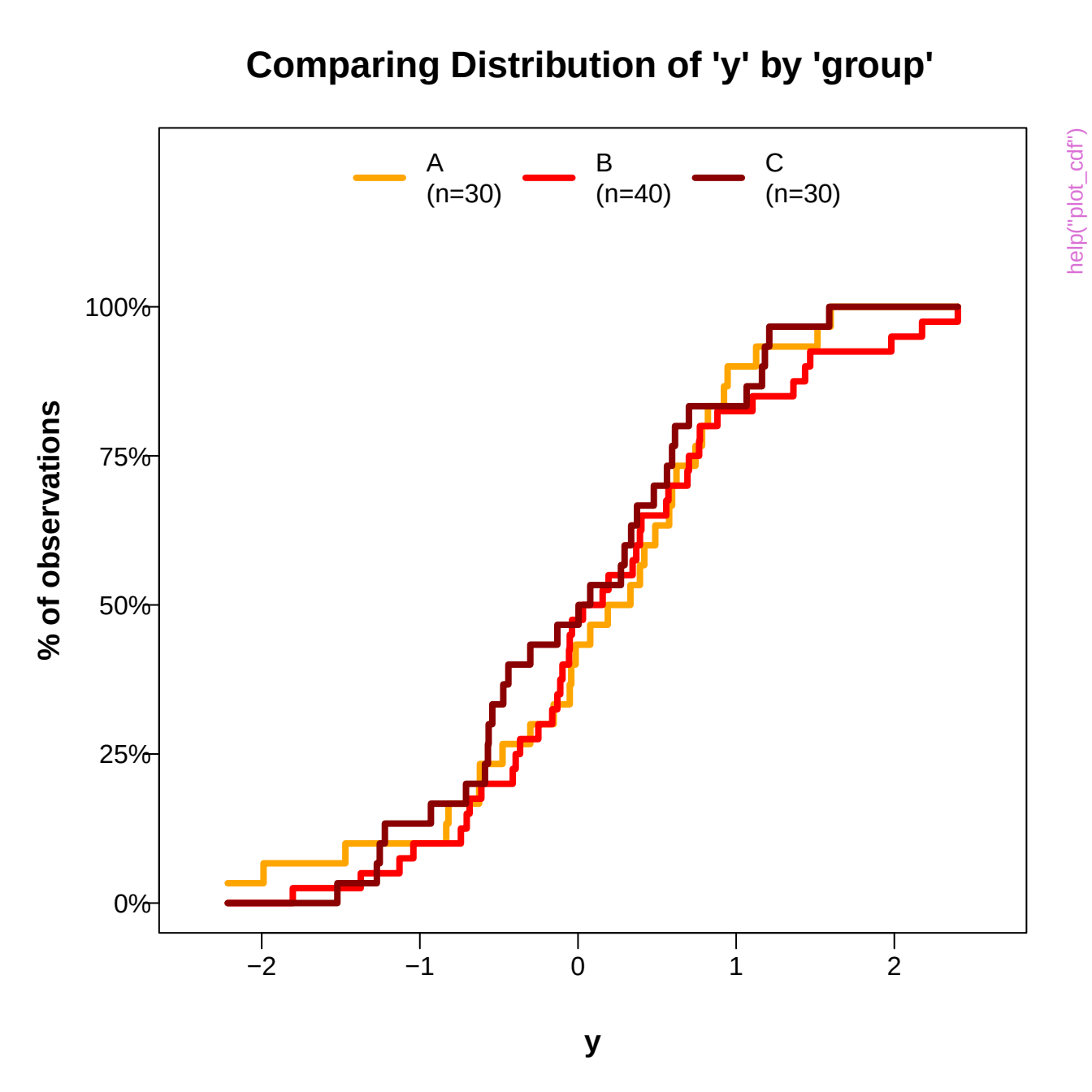

GAM Simple Slopes
Focal predictor ('treat')
Control
Treated
y
12
10
8
6
Number of Observations
44
48
40
29
39
34
42
35
41
48
0
2
4
6
8
10
Focal predictor ('treat')
Treated − Control
Marginal effect of treat
3
2
1
0
Number of Observations
44
48
40
29
39
34
42
35
41
48
0
2
4
6
8
10
help("interprobe_sim")

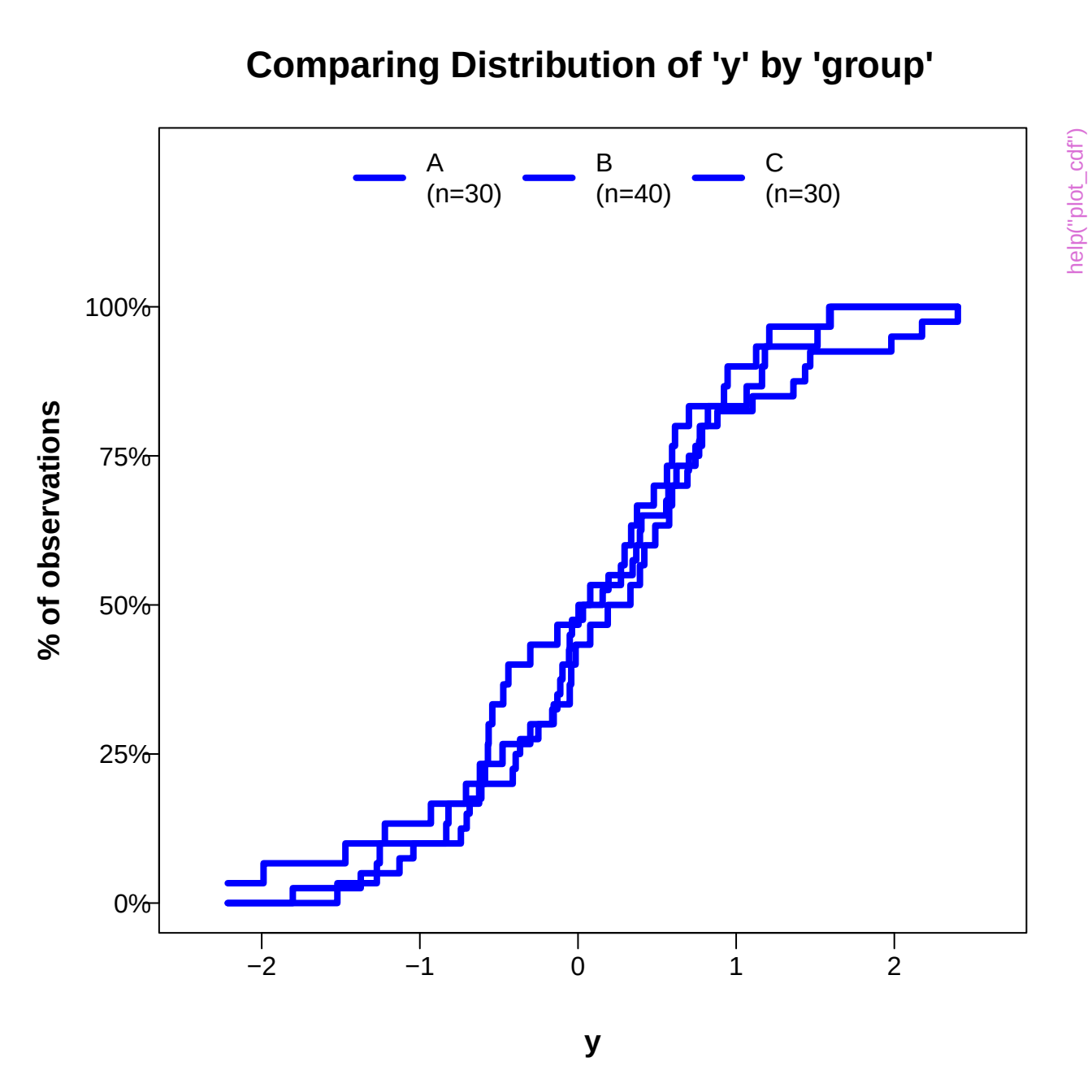

Scatter GAM − x & y
help("lm2")
y
10
5
0
Frequency
20
15
10
5
0
−3
−2
−1
0
1
2
3
x

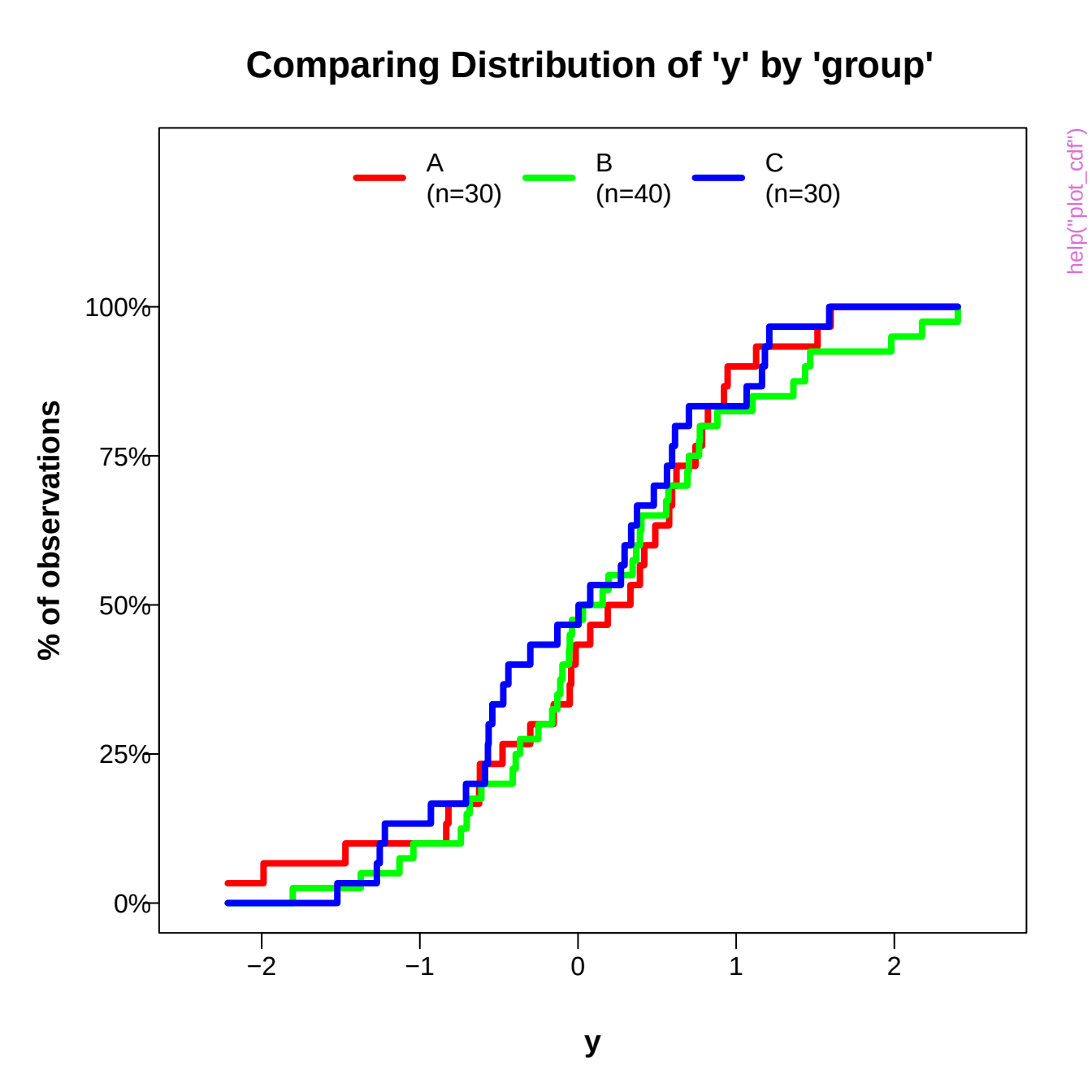

Distribution of y
(N=200)
x=1.6
9
Frequency
y
0
2
4
6
8
10
0
10
20
30
40
50
60
70
80
90
100
help("lm2")

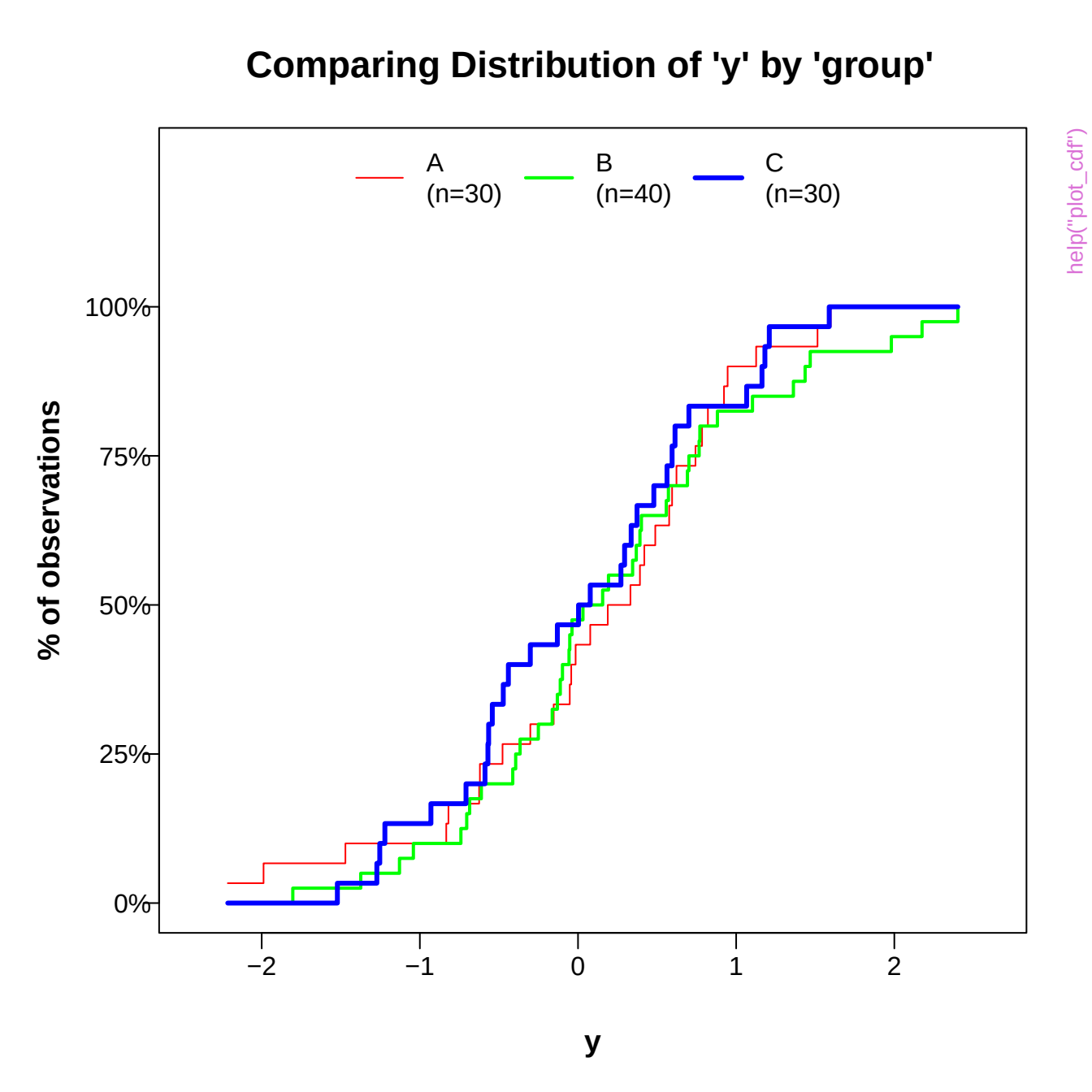

Distribution of x2
(N=200)
Frequency
80
60
40
20
0
76
49
44
17
12
1
1
−2
1
50
x2
help("lm2")

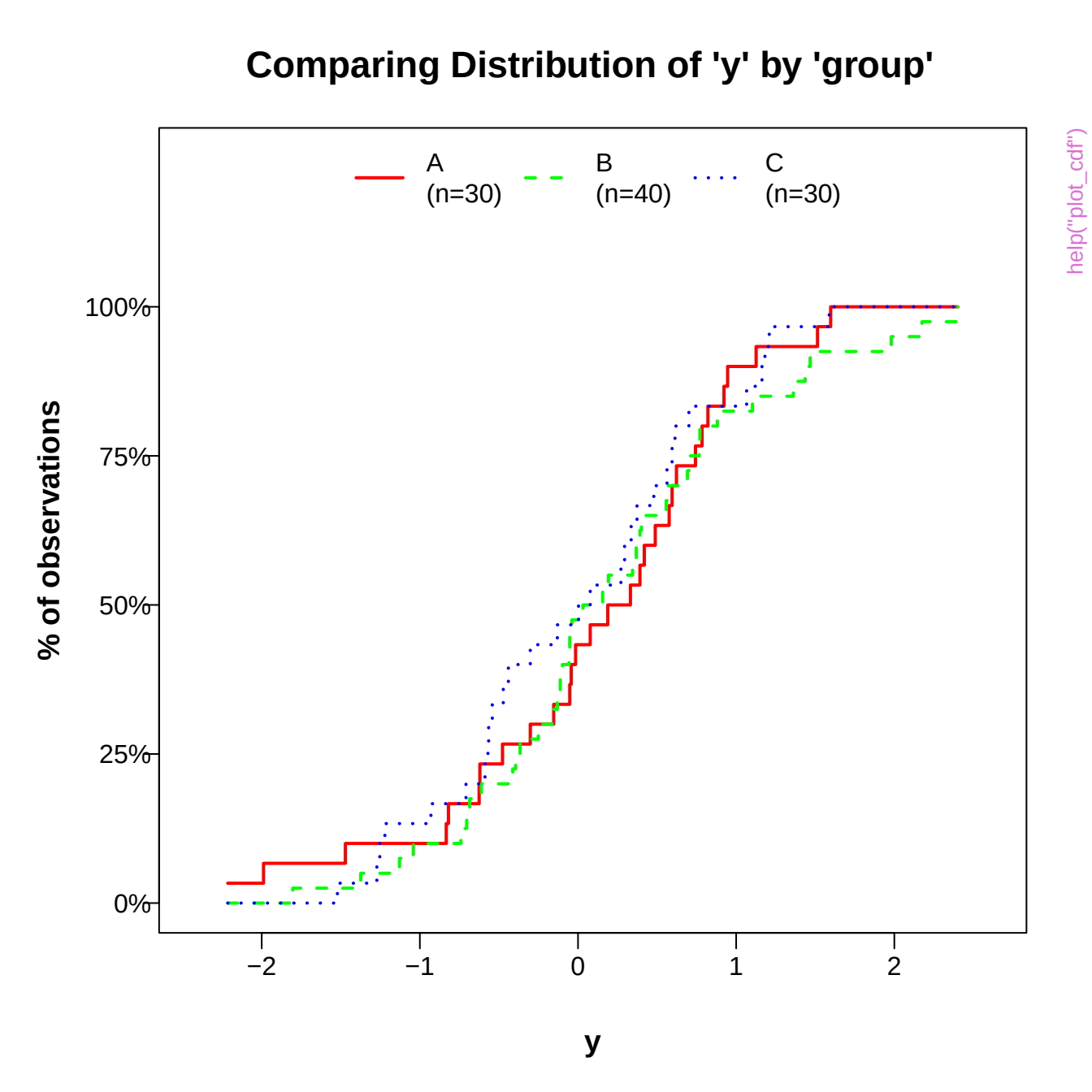

Distribution of 'y'
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

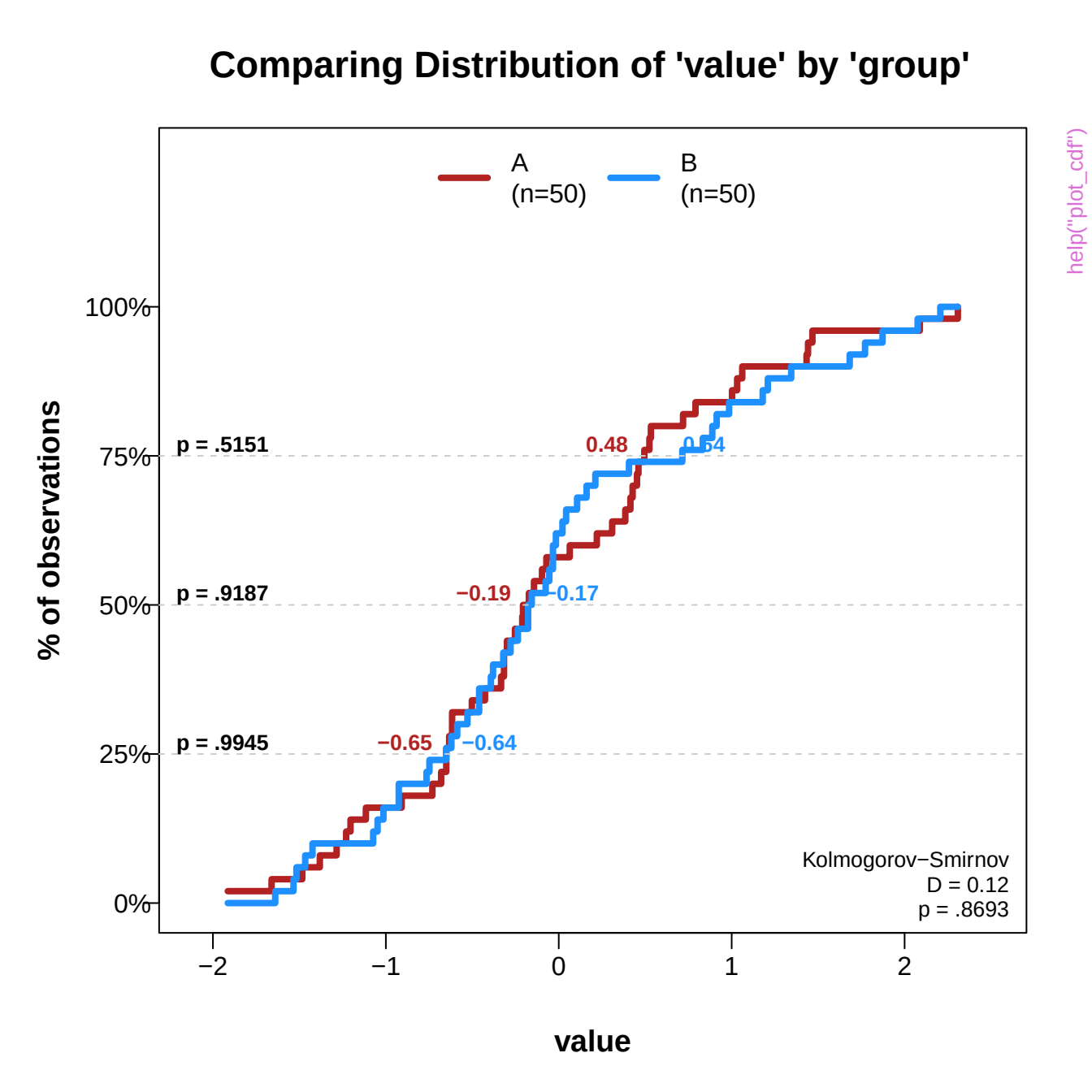

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

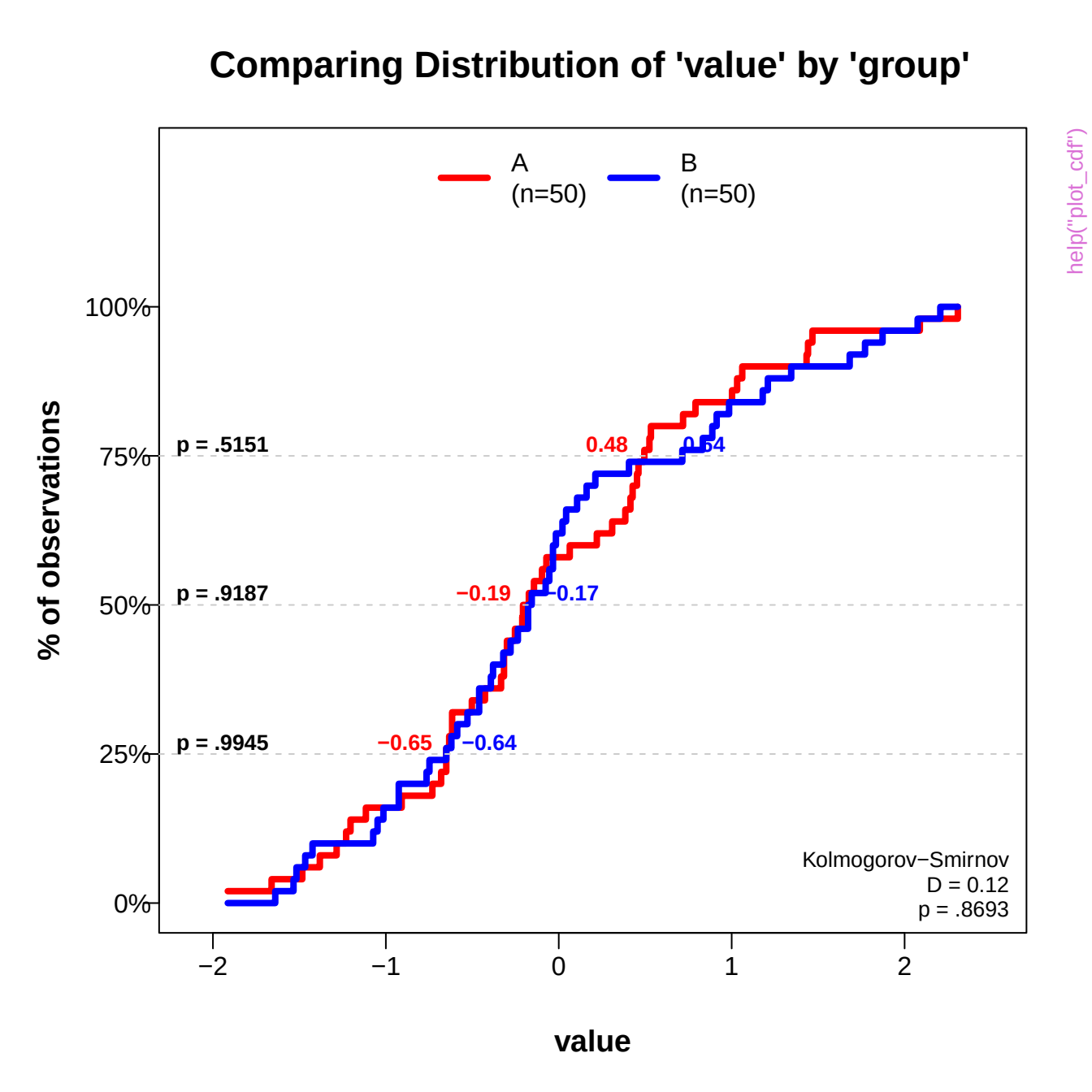

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

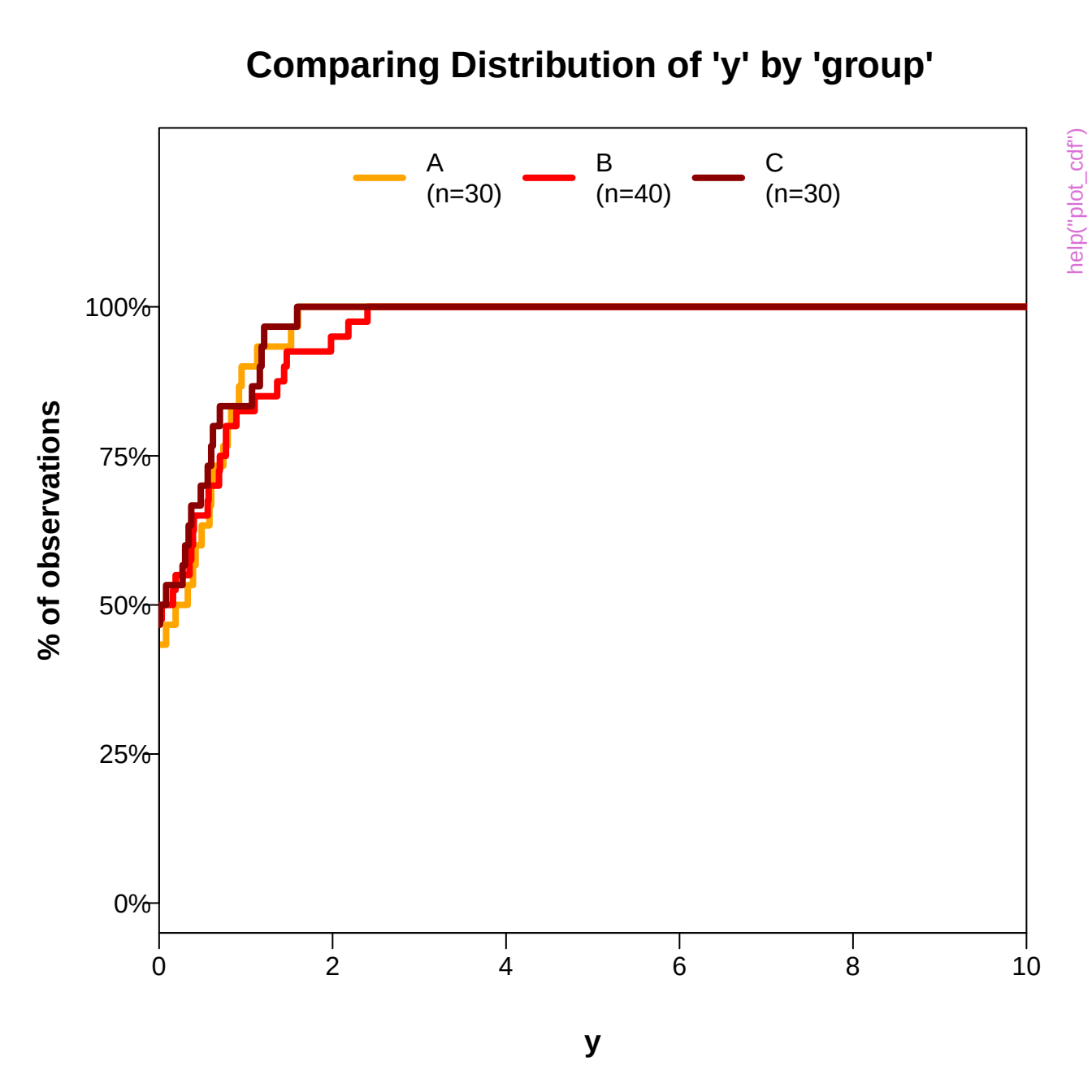

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

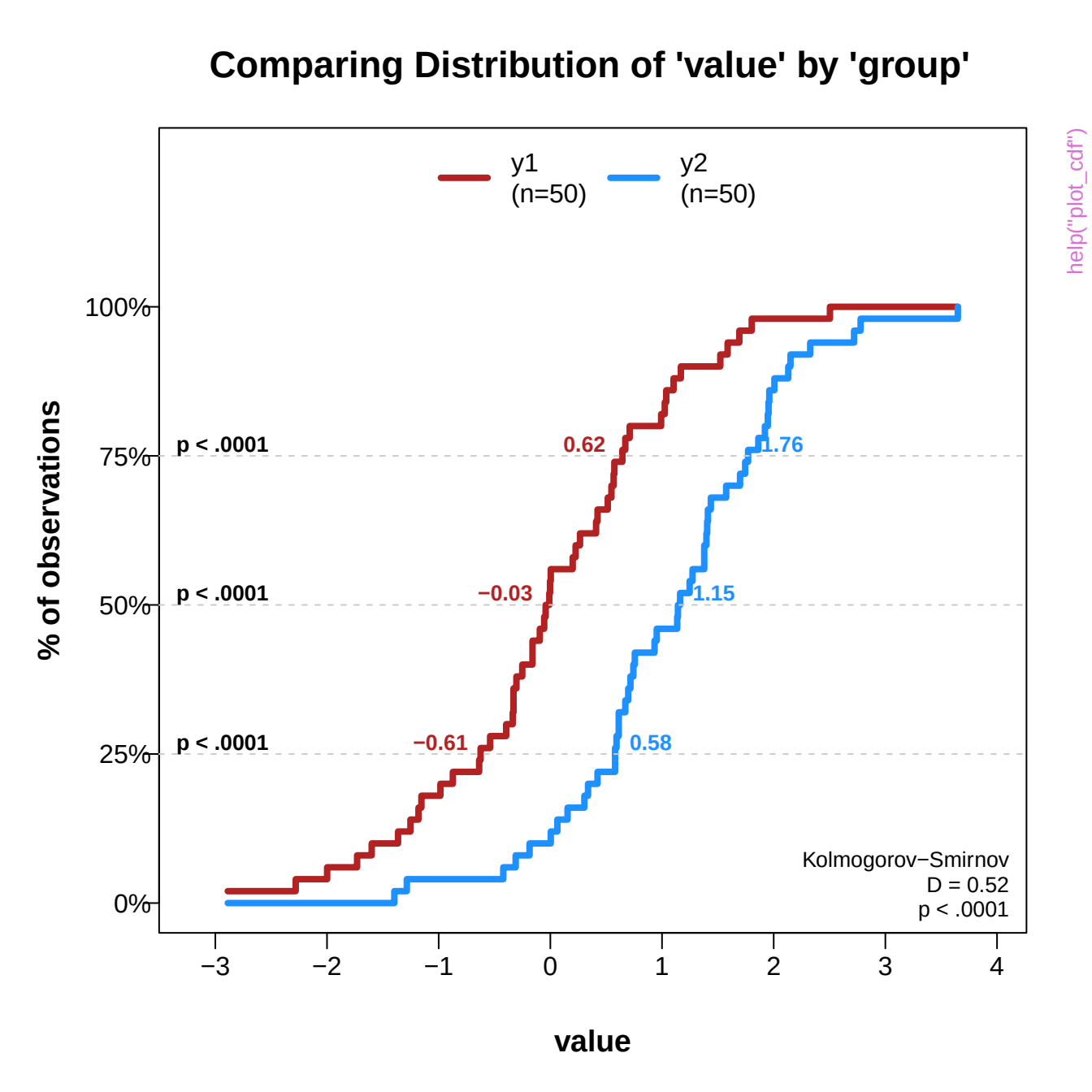

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

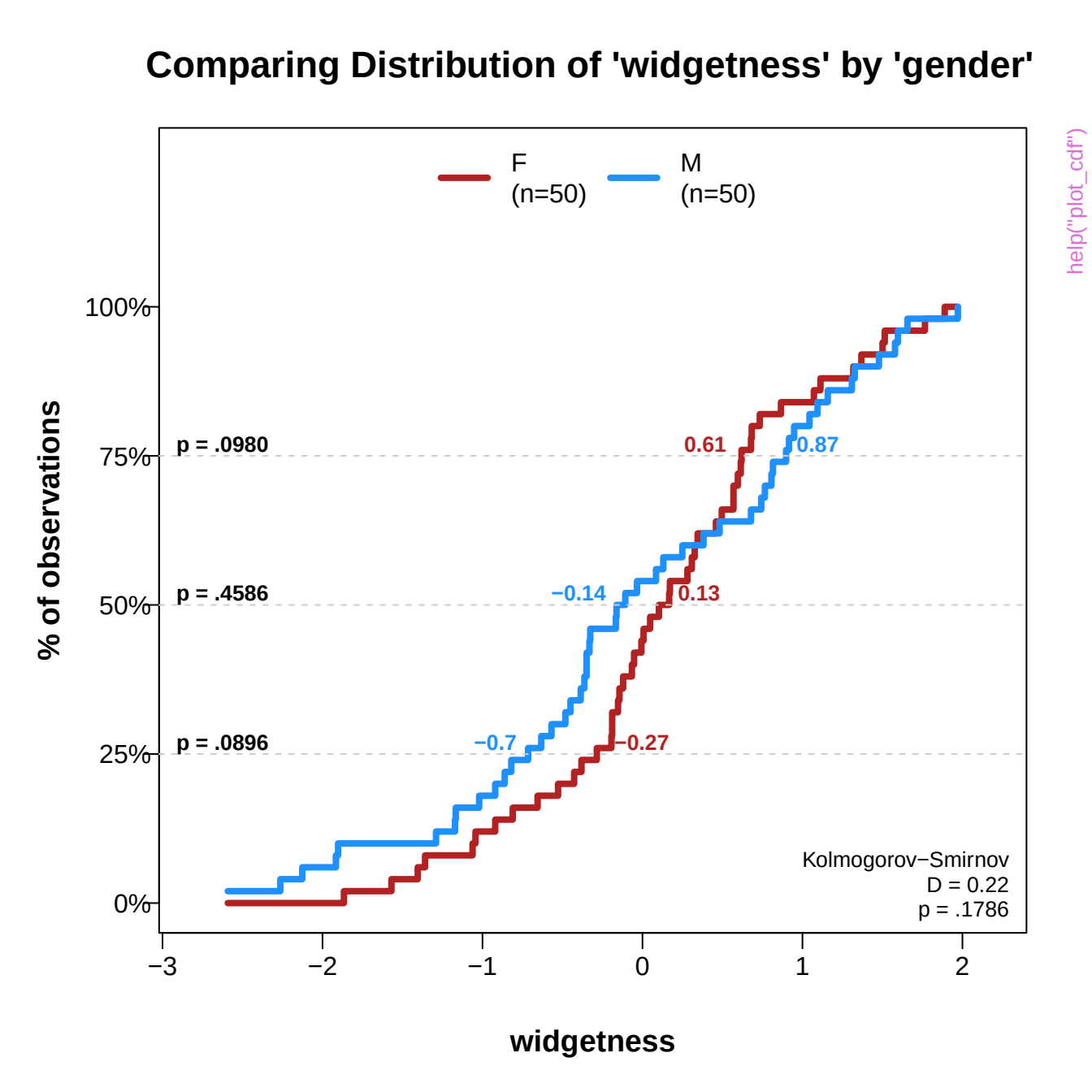

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

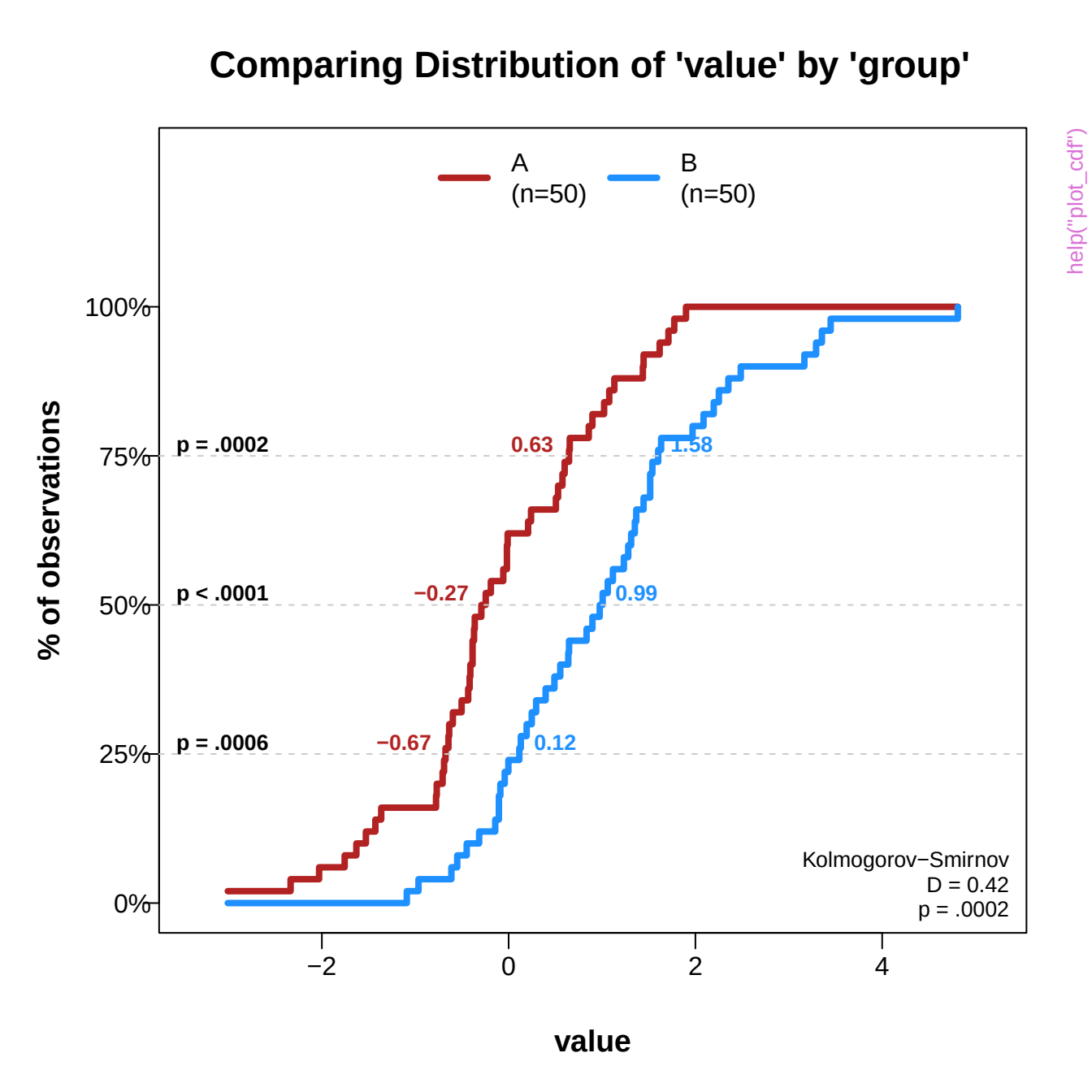

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
help("plot_cdf")
100%
75%
50%
25%
0%
% of observations
p = .5151
p = .9187
p = .9945
0.48
0.54
−0.19
−0.17
−0.65
−0.64
Kolmogorov−Smirnov
D = 0.12
p = .8693
−2
−1
0
1
2
value

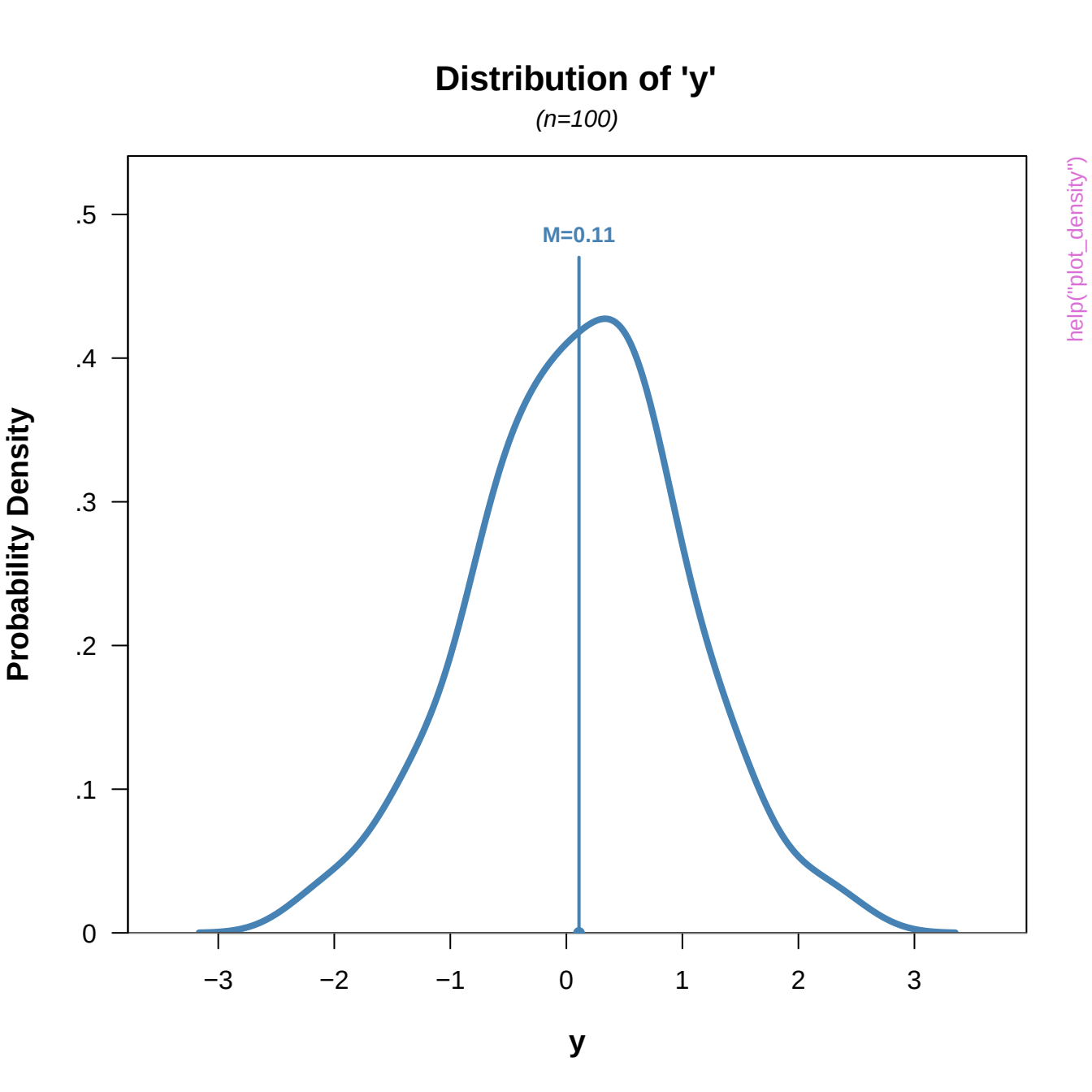

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
help("plot_cdf")
100%
75%
50%
25%
0%
% of observations
p = .5151
p = .9187
p = .9945
0.48
0.64
−0.19
−0.17
−0.65
−0.64
Kolmogorov−Smirnov
D = 0.12
p = .8693
−2
−1
0
1
2
value

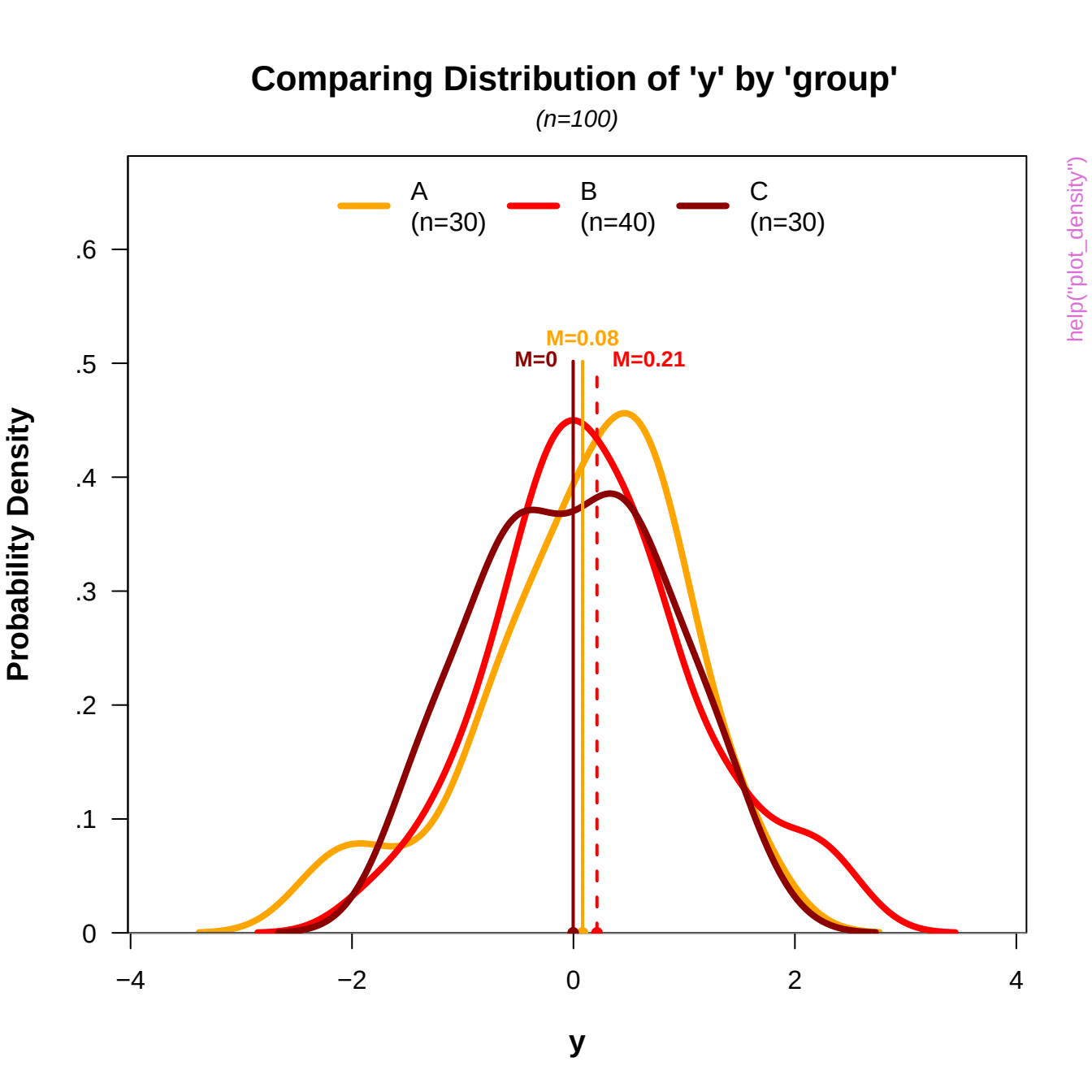

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
0
2
4
6
8
10
y
help("plot_cdf")

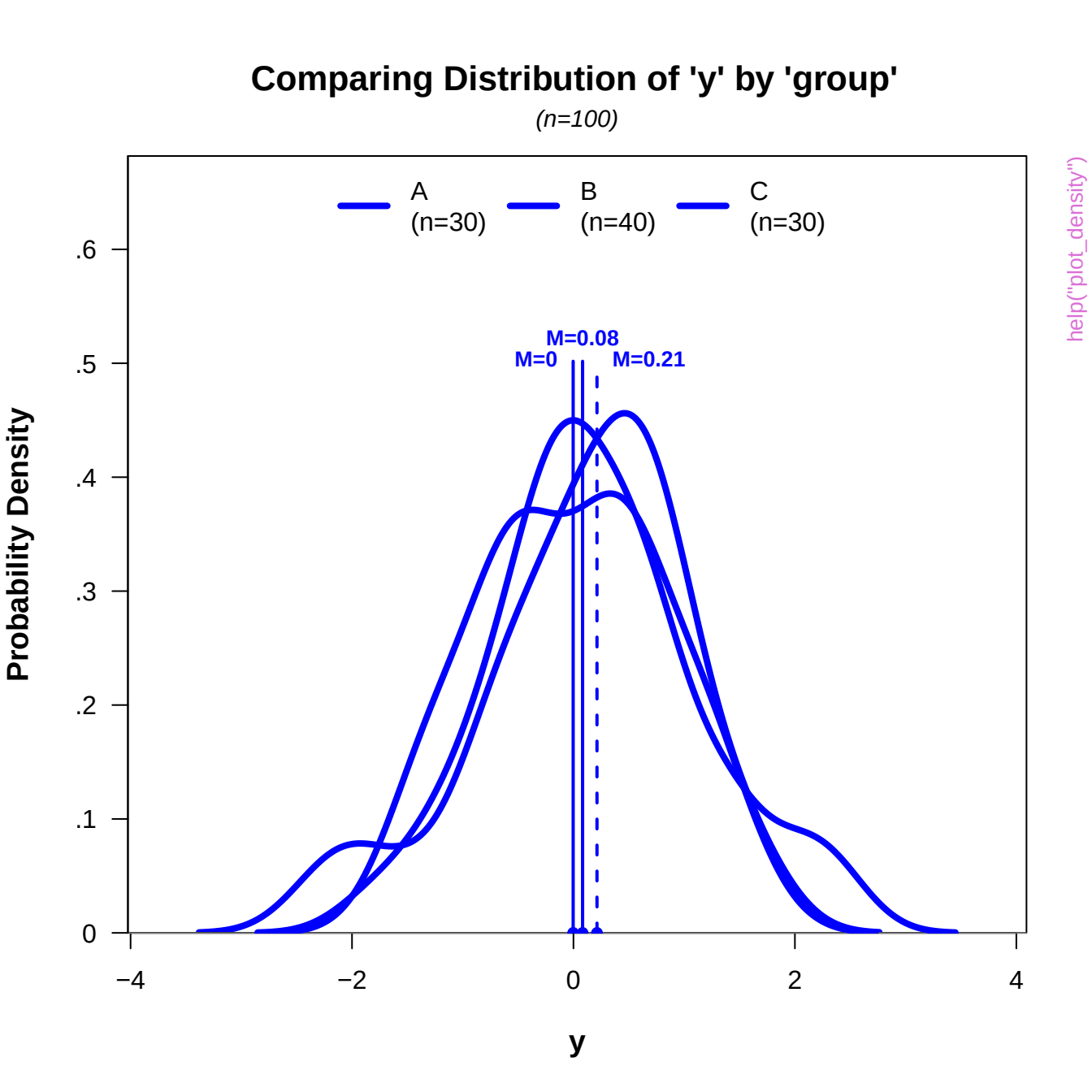

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=50)
help("plot_cdf")
100%
75%
50%
25%
0%
% of observations
p < .0001
p < .0001
p < .0001
0.62
1.76
−0.03
1.15
−0.61
0.58
Kolmogorov−Smirnov
D = 0.52
p < .0001
−3
−2
−1
0
1
2
3
4
value

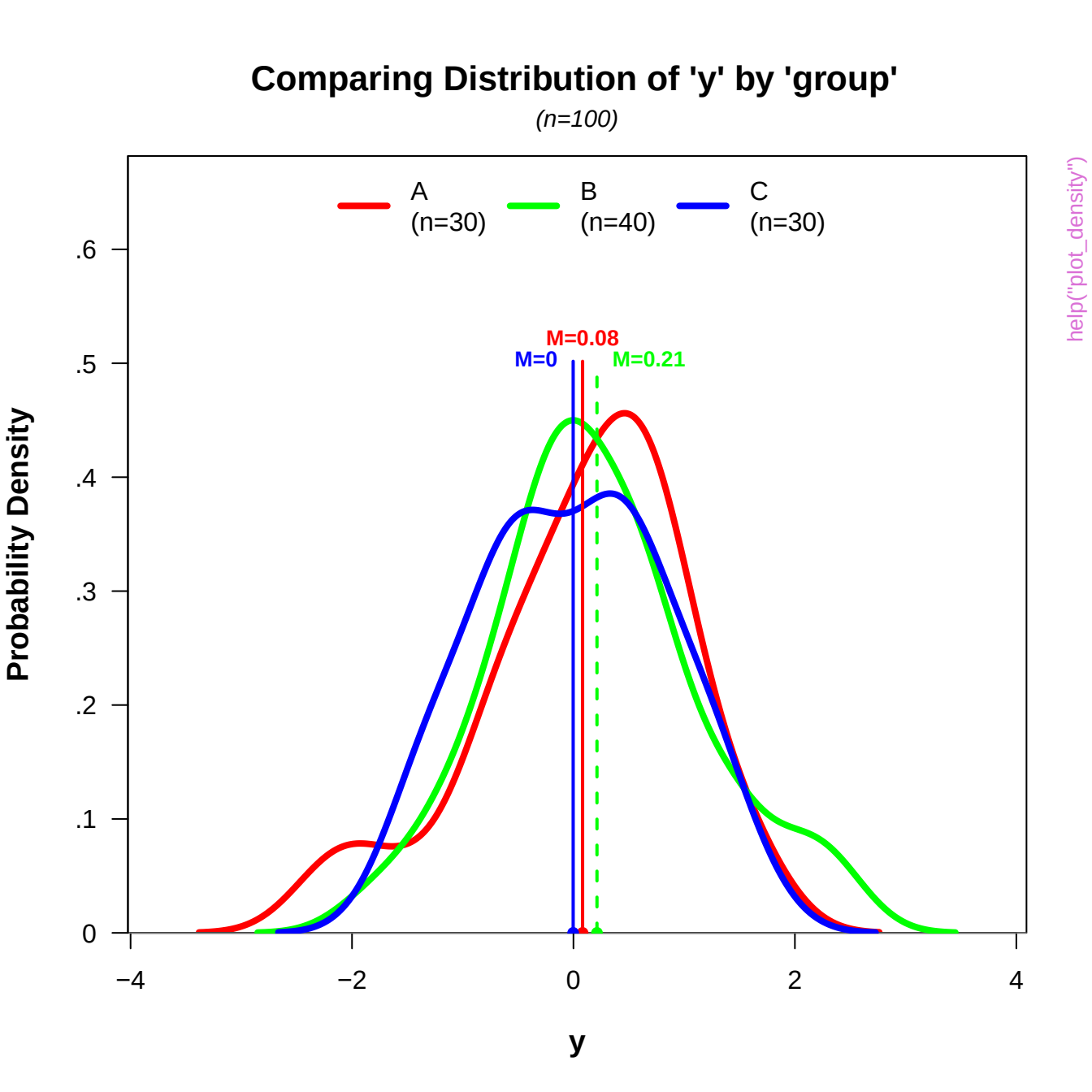

Comparing Distribution of 'widgetness' by 'gender'
F
(n=50)
M
(n=50)
help("plot_cdf")
100%
75%
50%
25%
0%
% of observations
p = .0980
p = .4586
p = .0896
0.61
0.87
−0.14
0.13
−0.7
−0.27
Kolmogorov−Smirnov
D = 0.22
p = .1786
−3
−2
−1
0
1
2
widgetness

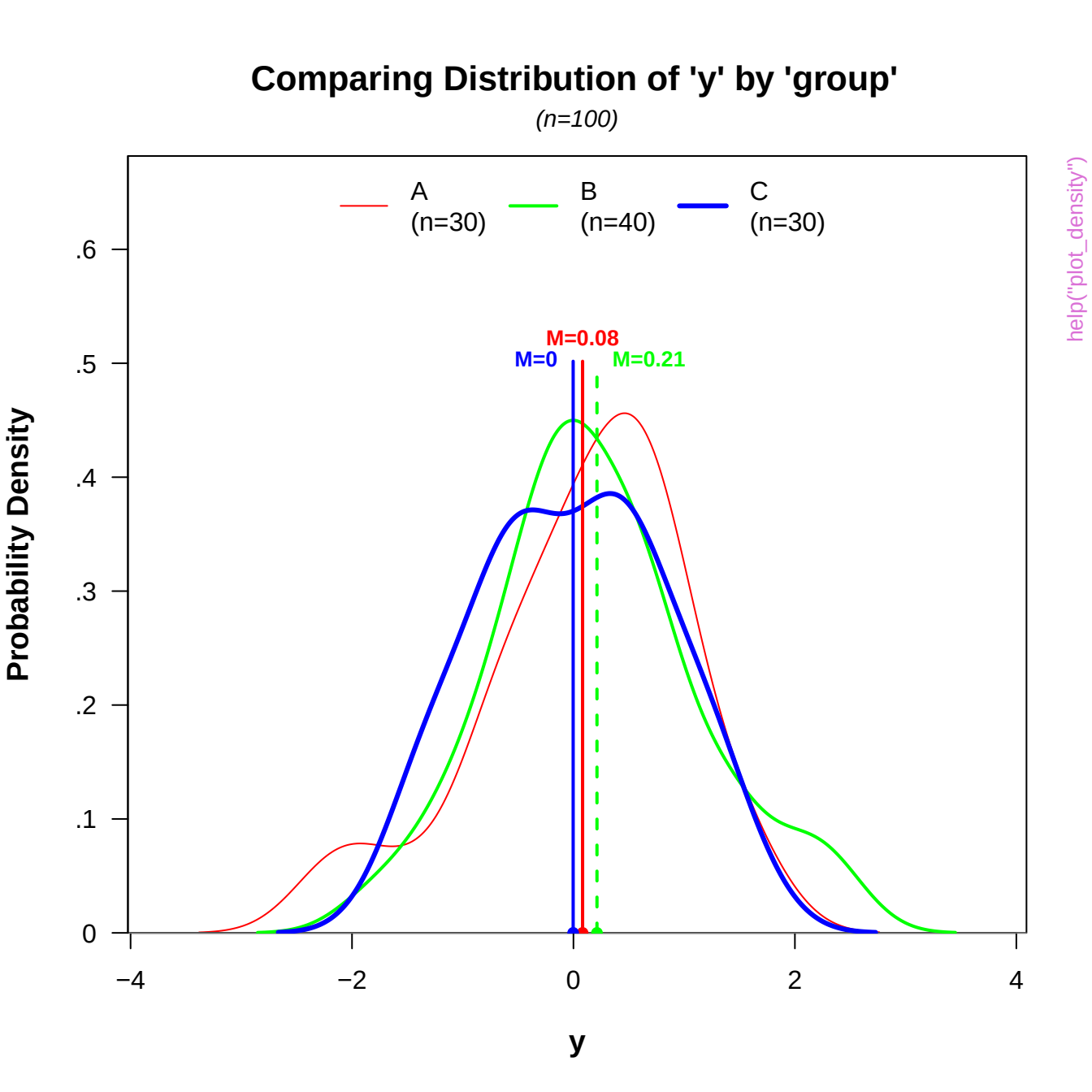

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
help("plot_cdf")
100%
75%
50%
25%
0%
% of observations
p = .0002
p < .0001
p = .0006
0.63
1.58
−0.27
0.99
−0.67
0.12
Kolmogorov−Smirnov
D = 0.42
p = .0002
−2
0
2
4
value

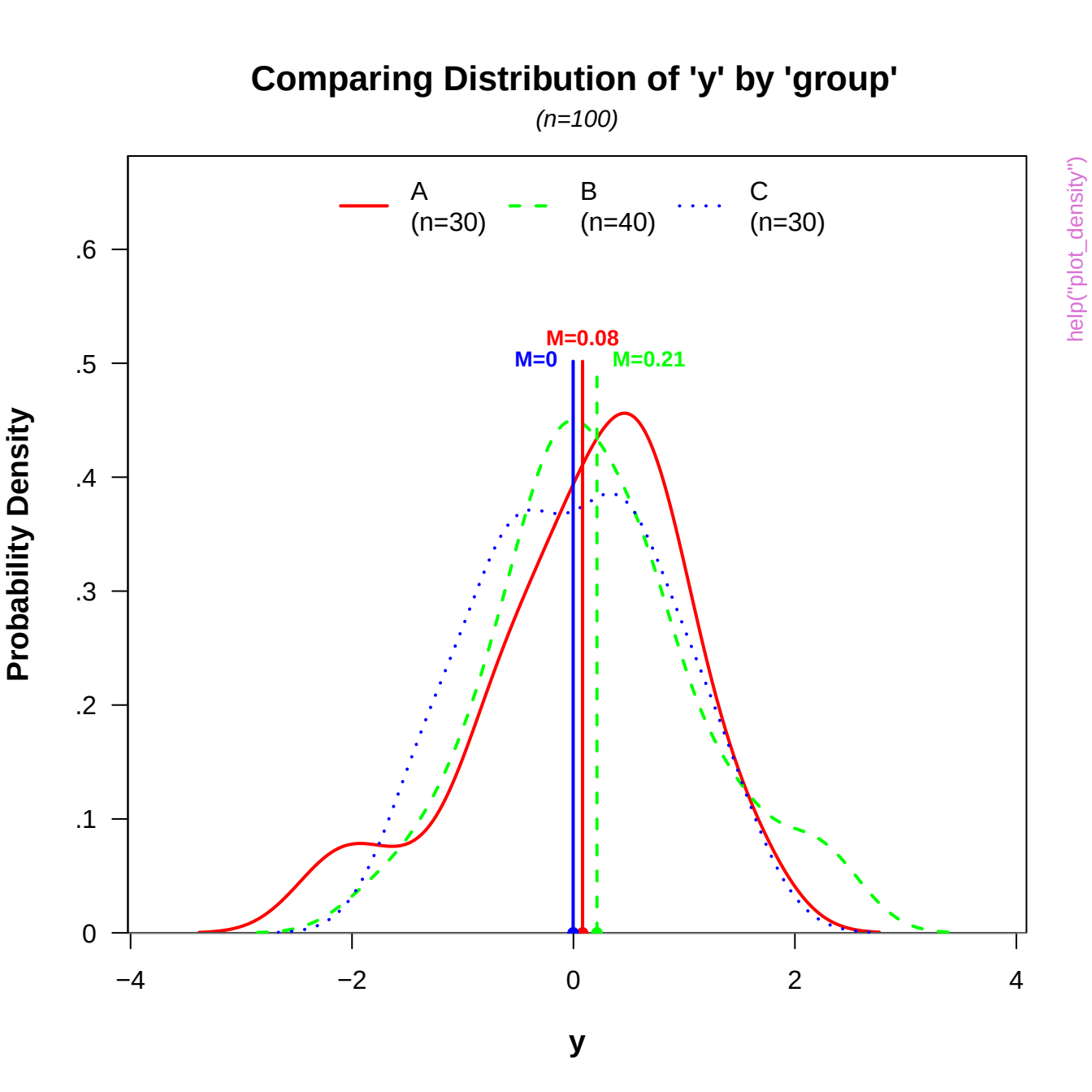

Distribution of 'y'
(n=100)
M=0.11
Probability Density
.5
.4
.3
.2
.1
0
−3
−2
−1
0
1
2
3
y
help("plot_density")

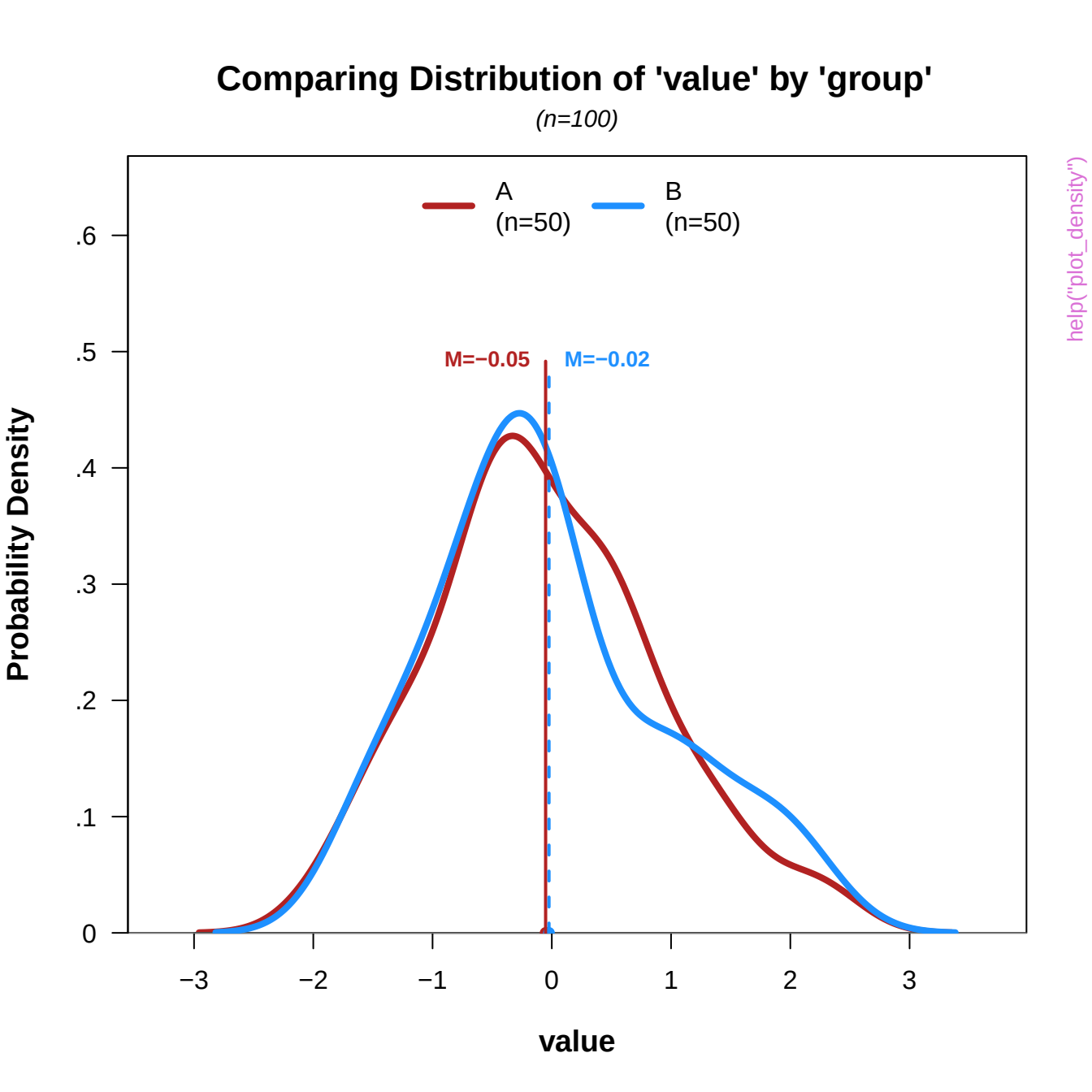

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
0
.1
.2
.3
.4
.5
.6
−4
−2
0
2
4
y
help("plot_density")

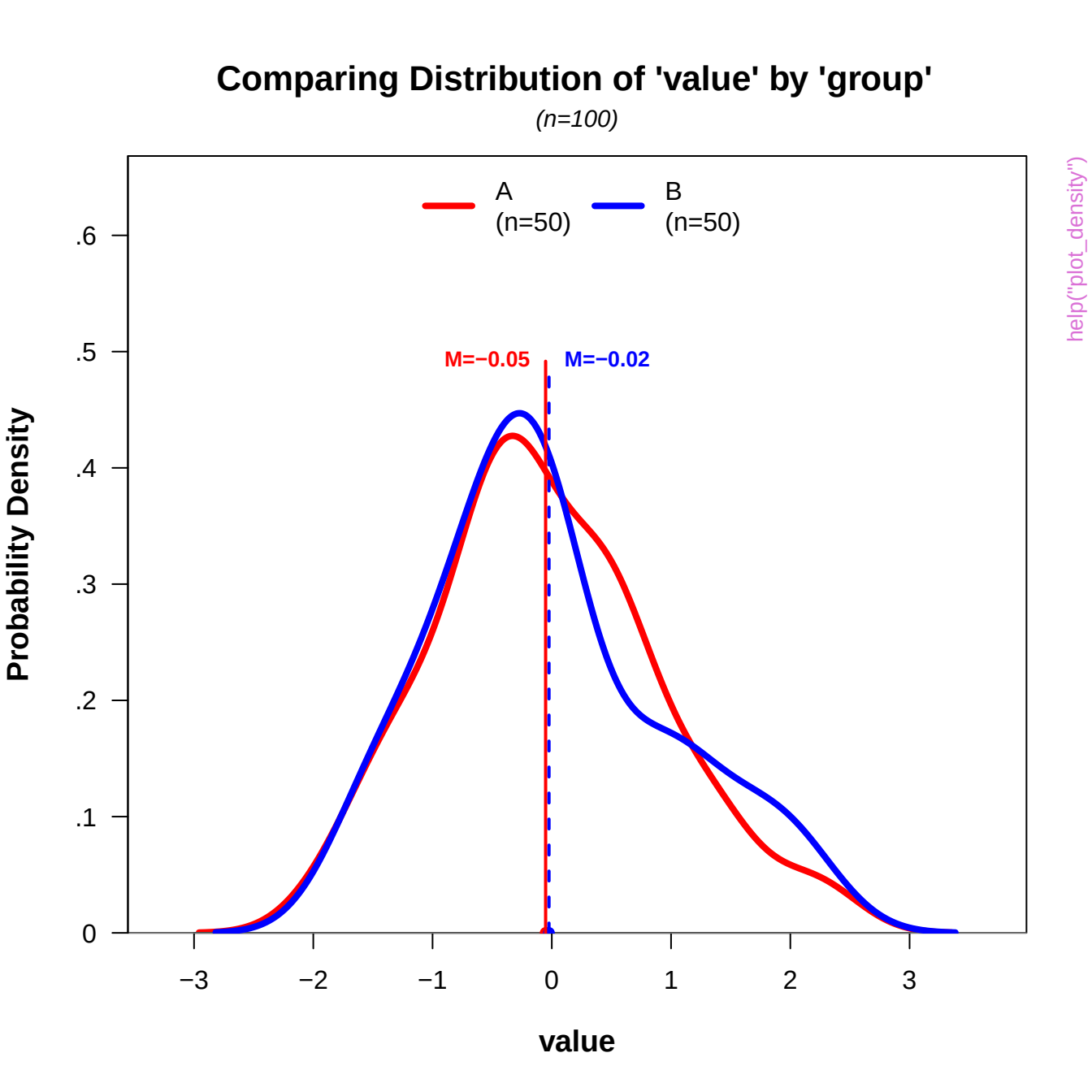

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
0
.1
.2
.3
.4
.5
.6
−4
−2
0
2
4
y
help("plot_density")

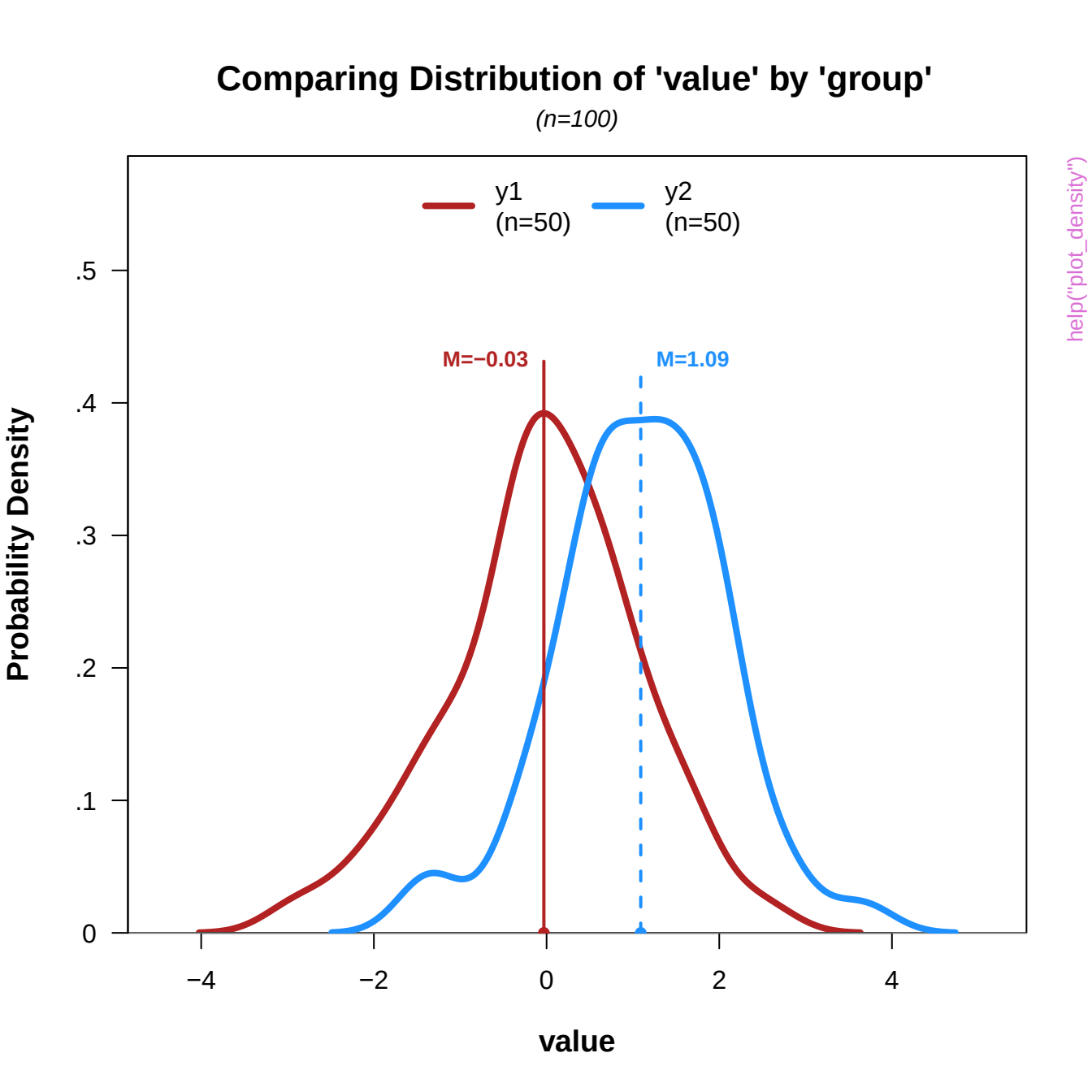

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y
help("plot_density")

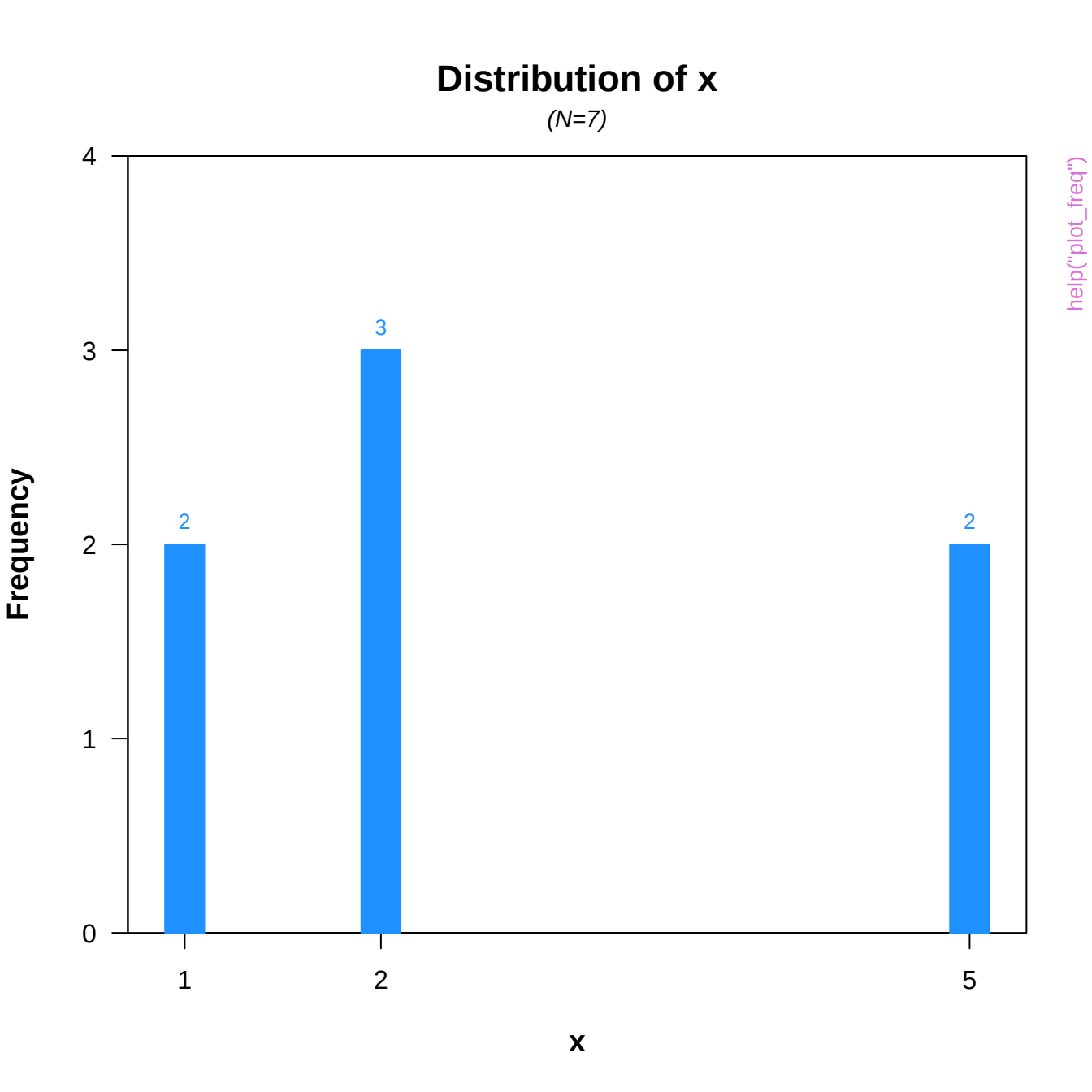

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y
help("plot_density")

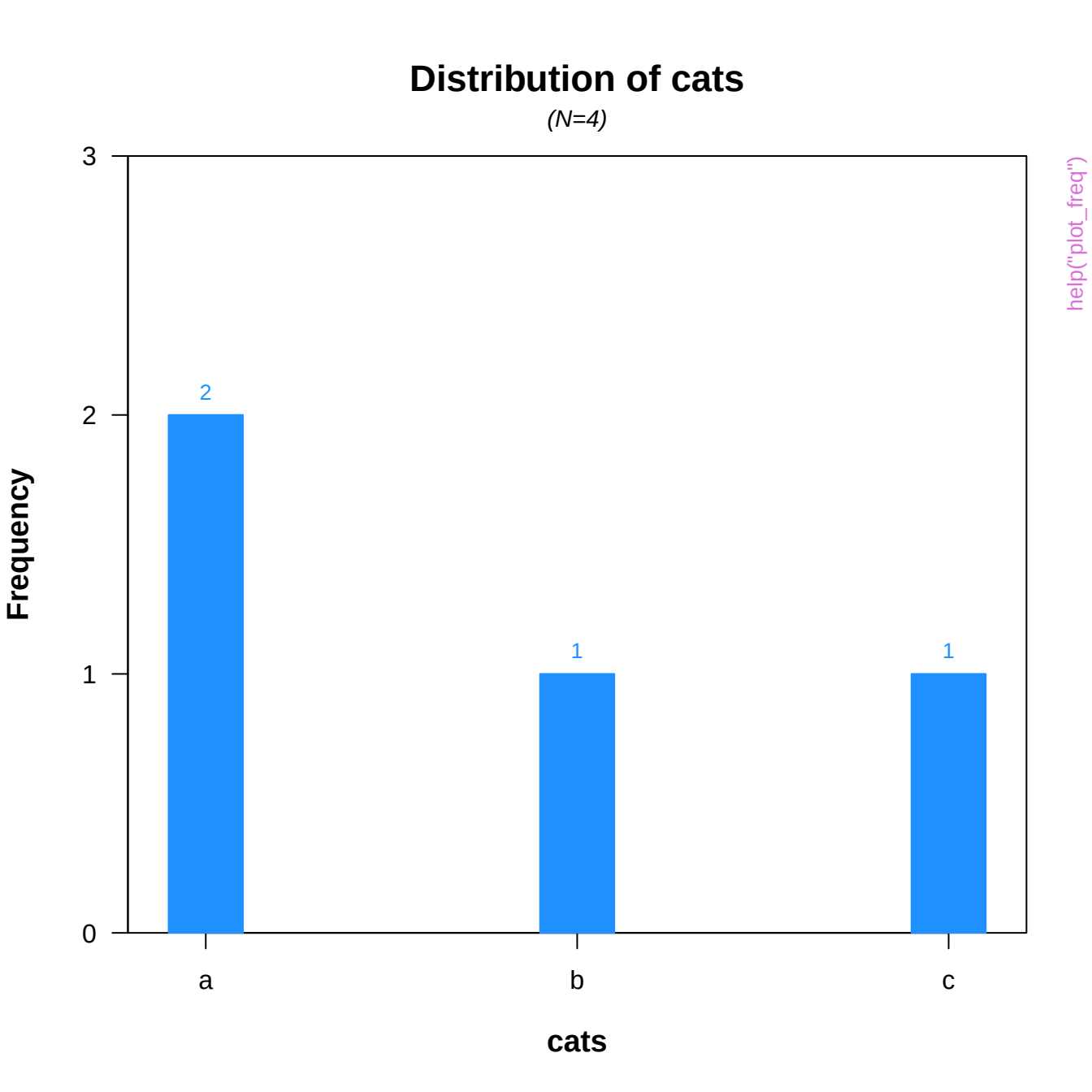

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
0
.1
.2
.3
.4
.5
.6
−4
−2
0
2
4
y
help("plot_density")

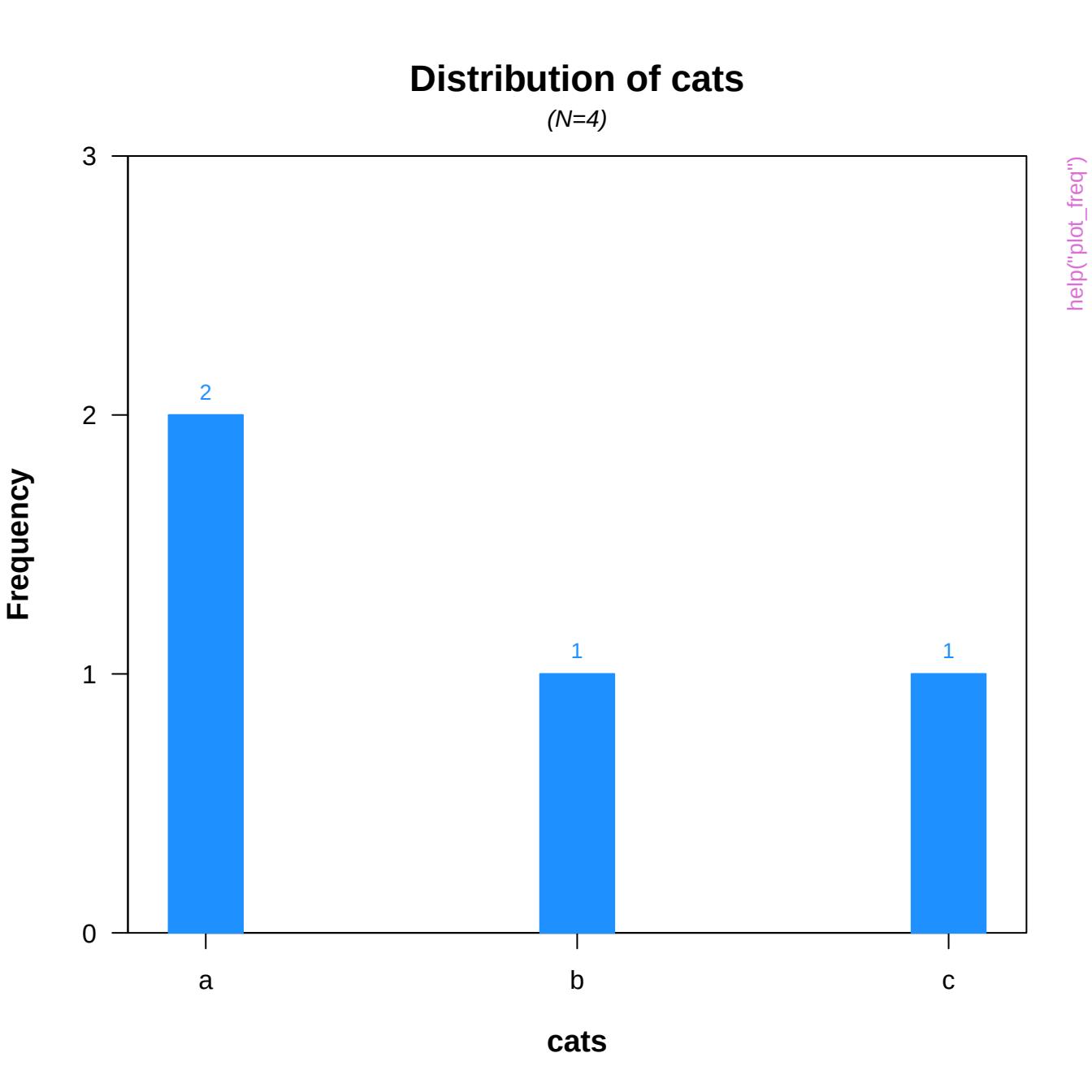

Comparing Distribution of 'value' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.05
M=−0.02
Probability Density
0
.1
.2
.3
.4
.5
.6
−3
−2
−1
0
1
2
3
value
help("plot_density")

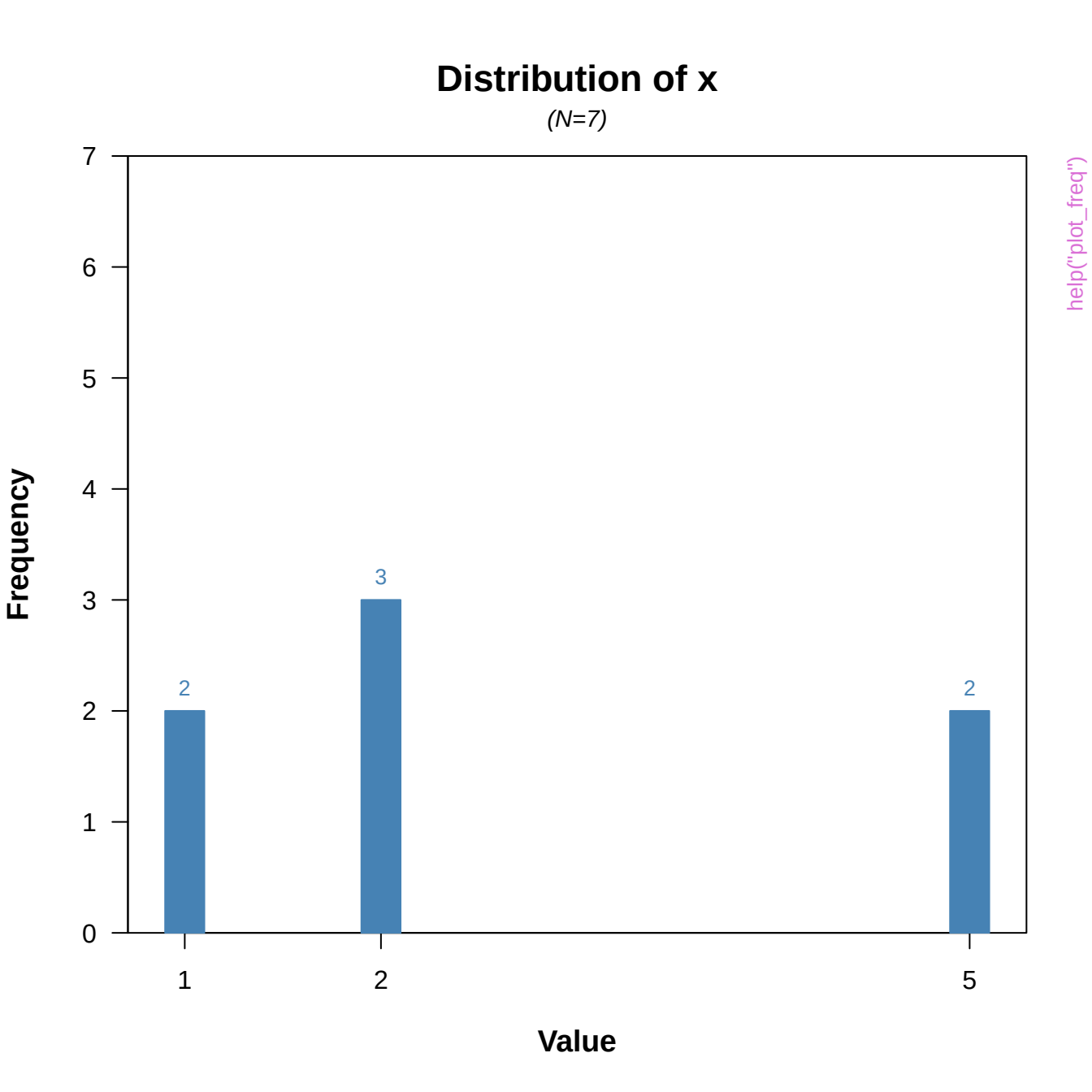

Comparing Distribution of 'value' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.05
M=−0.02
Probability Density
0
.1
.2
.3
.4
.5
.6
−3
−2
−1
0
1
2
3
value
help("plot_density")

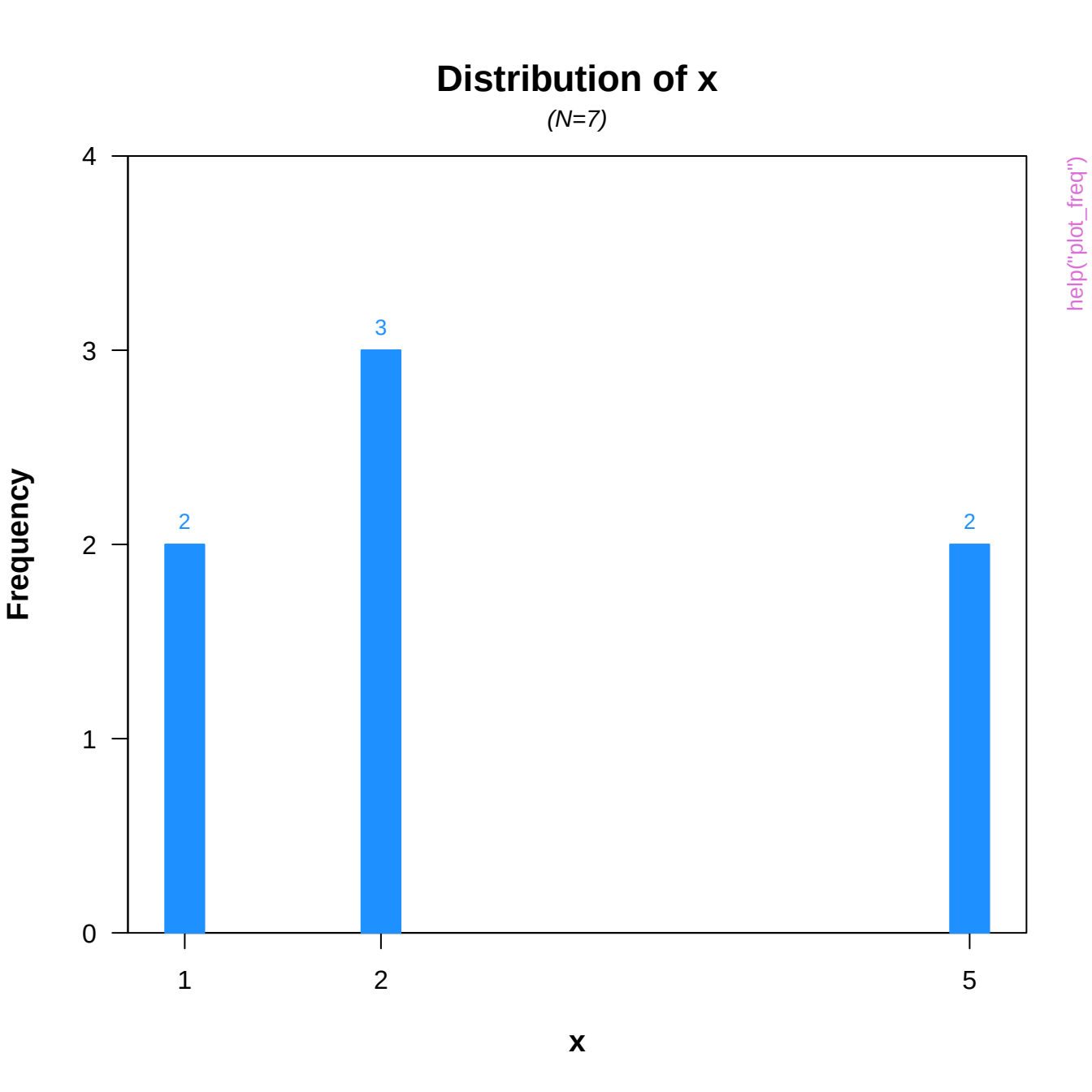

Comparing Distribution of 'value' by 'group'
(n=100)
y1
(n=50)
y2
(n=50)
M=−0.03
M=1.09
Probability Density
0
.1
.2
.3
.4
.5
−4
−2
0
2
4
value
help("plot_density")

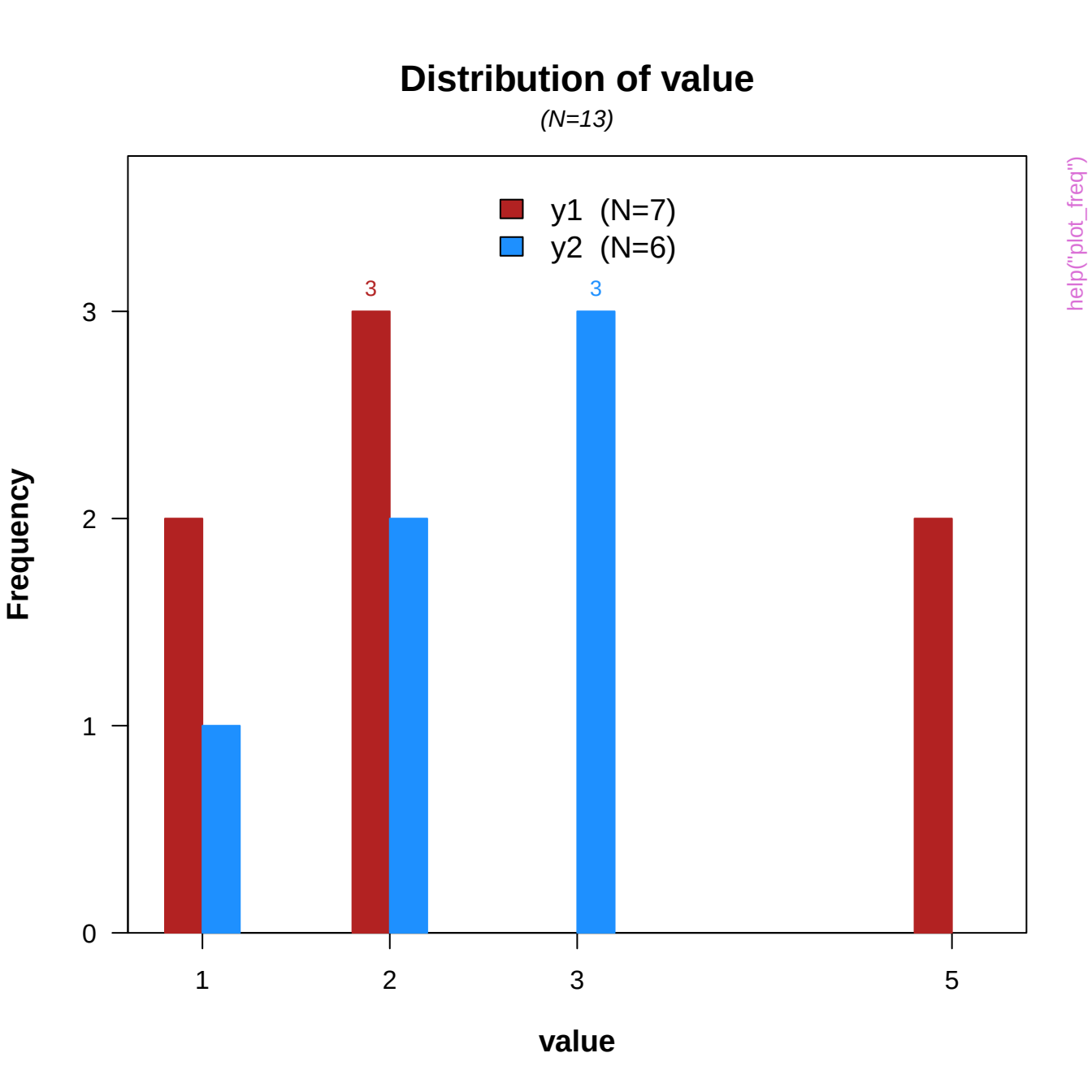

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
5
x
help("plot_freq")

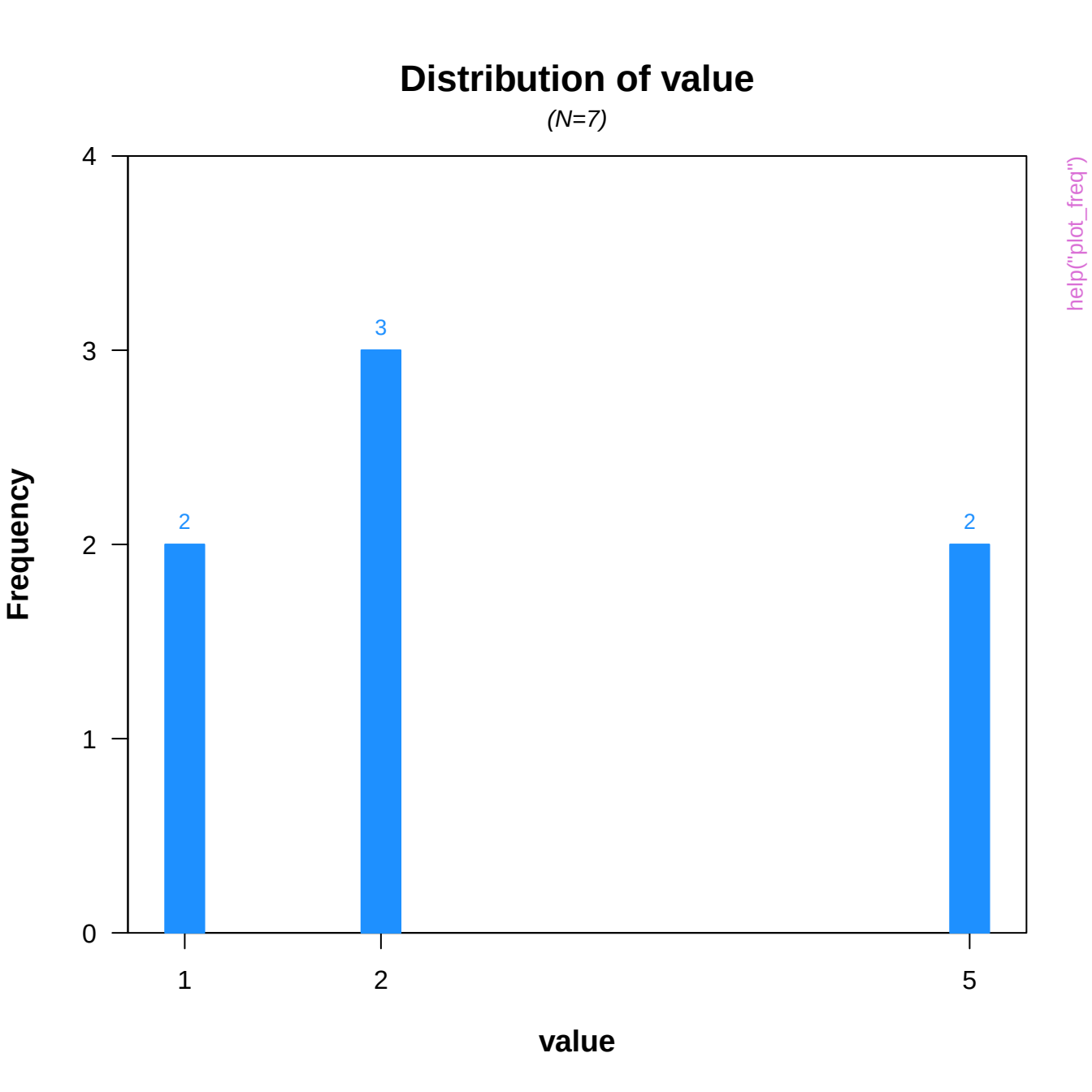

Distribution of cats
(N=4)
Frequency
3
2
1
0
2
1
1
a
b
c
cats
help("plot_freq")

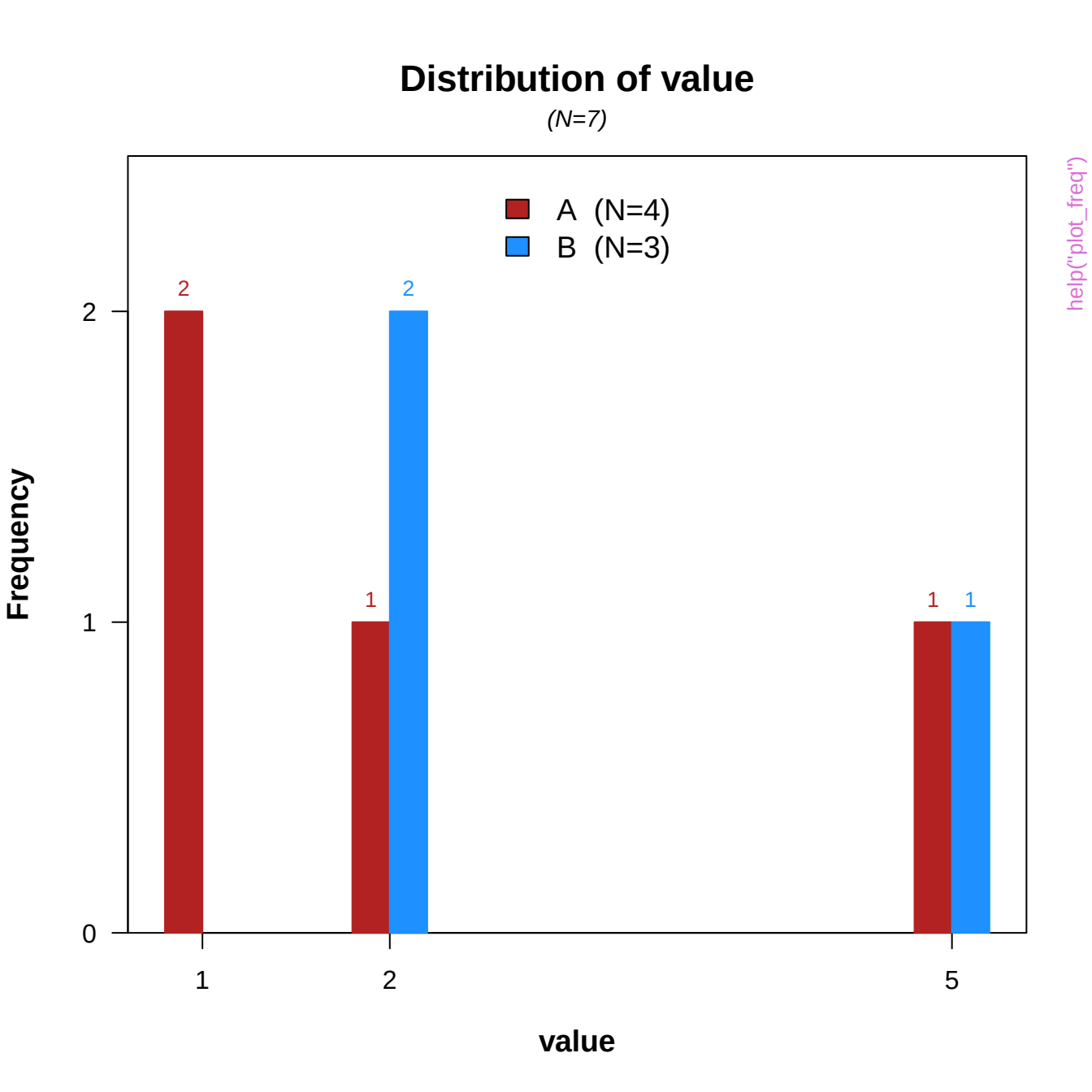

Distribution of cats
(N=4)
Frequency
3
2
1
0
2
1
1
a
b
c
cats
help("plot_freq")

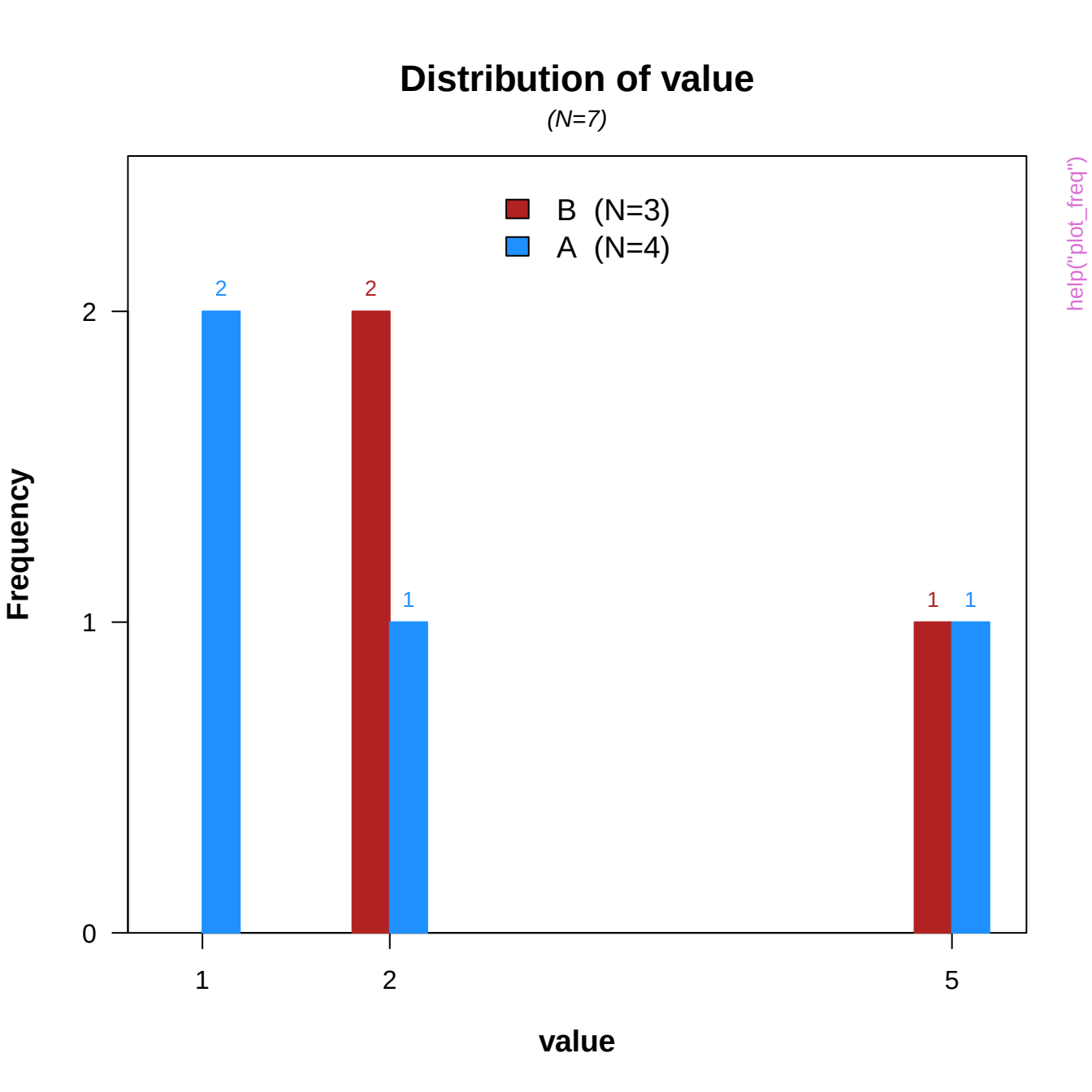

Distribution of x
(N=7)
Frequency
Value
help("plot_freq")
0
1
2
3
4
5
6
7
1
2
5
2
3
2

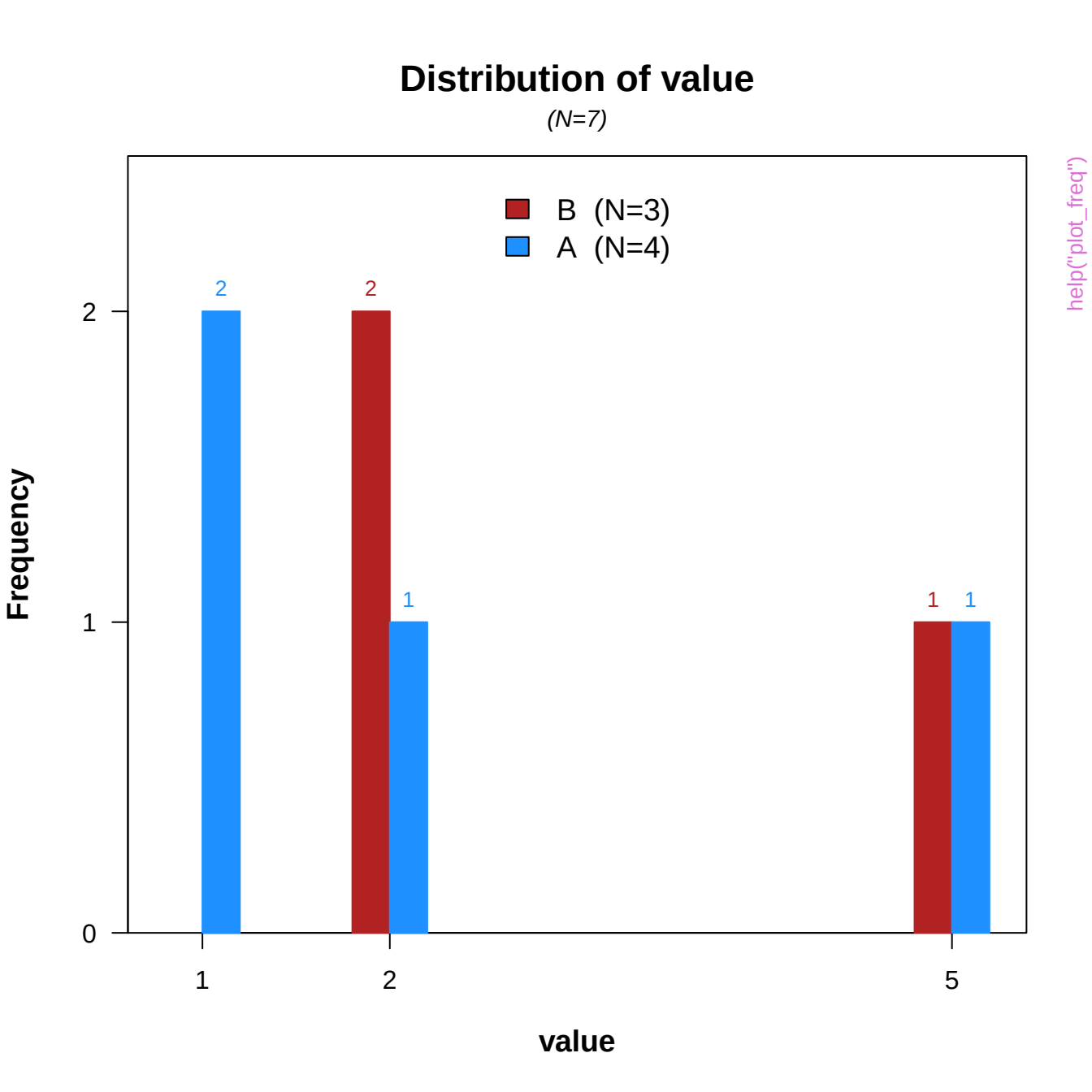

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
5
x
help("plot_freq")

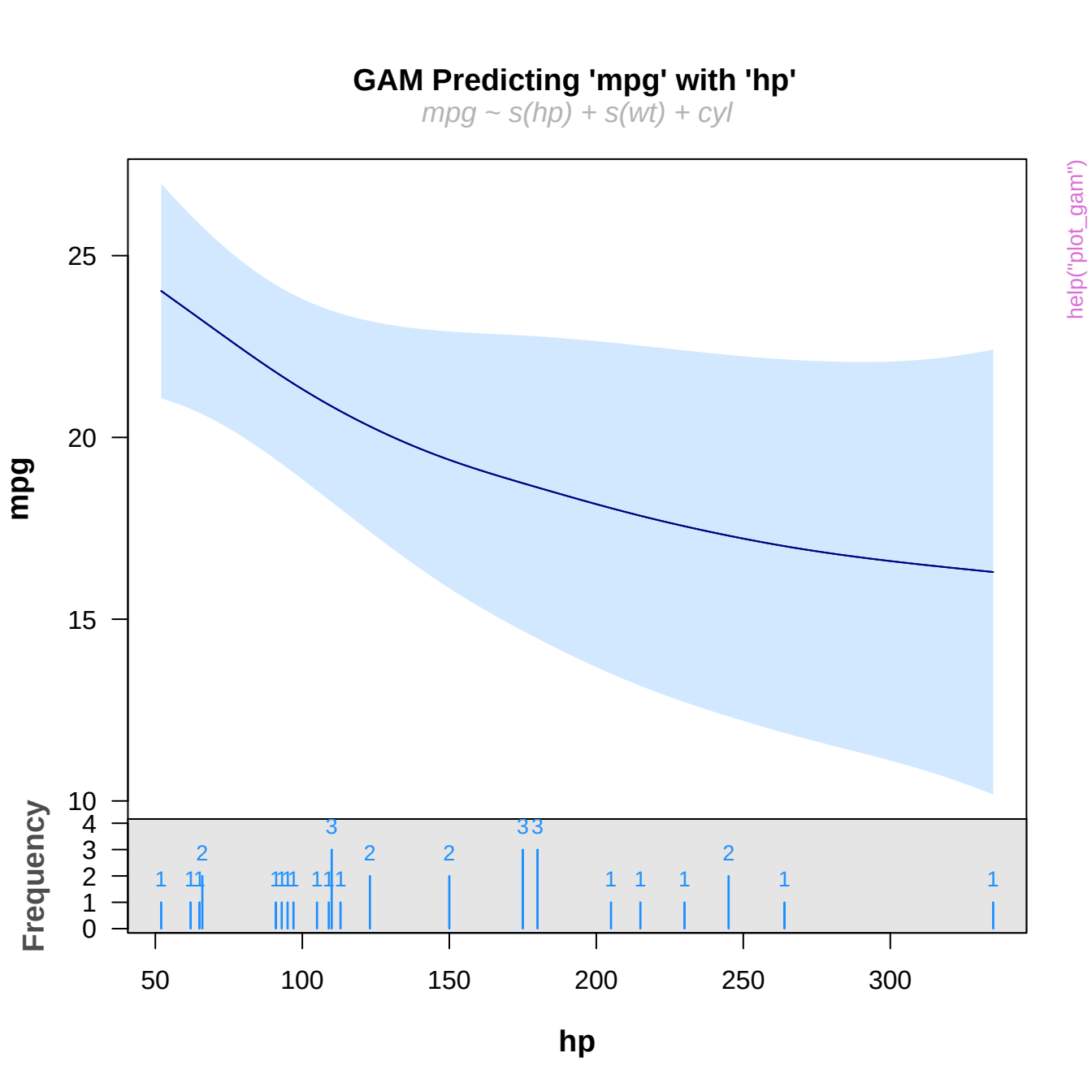

Distribution of value
(N=13)
y1 (N=7)
y2 (N=6)
3
3
Frequency
0
1
2
3
1
2
3
5
value
help("plot_freq")

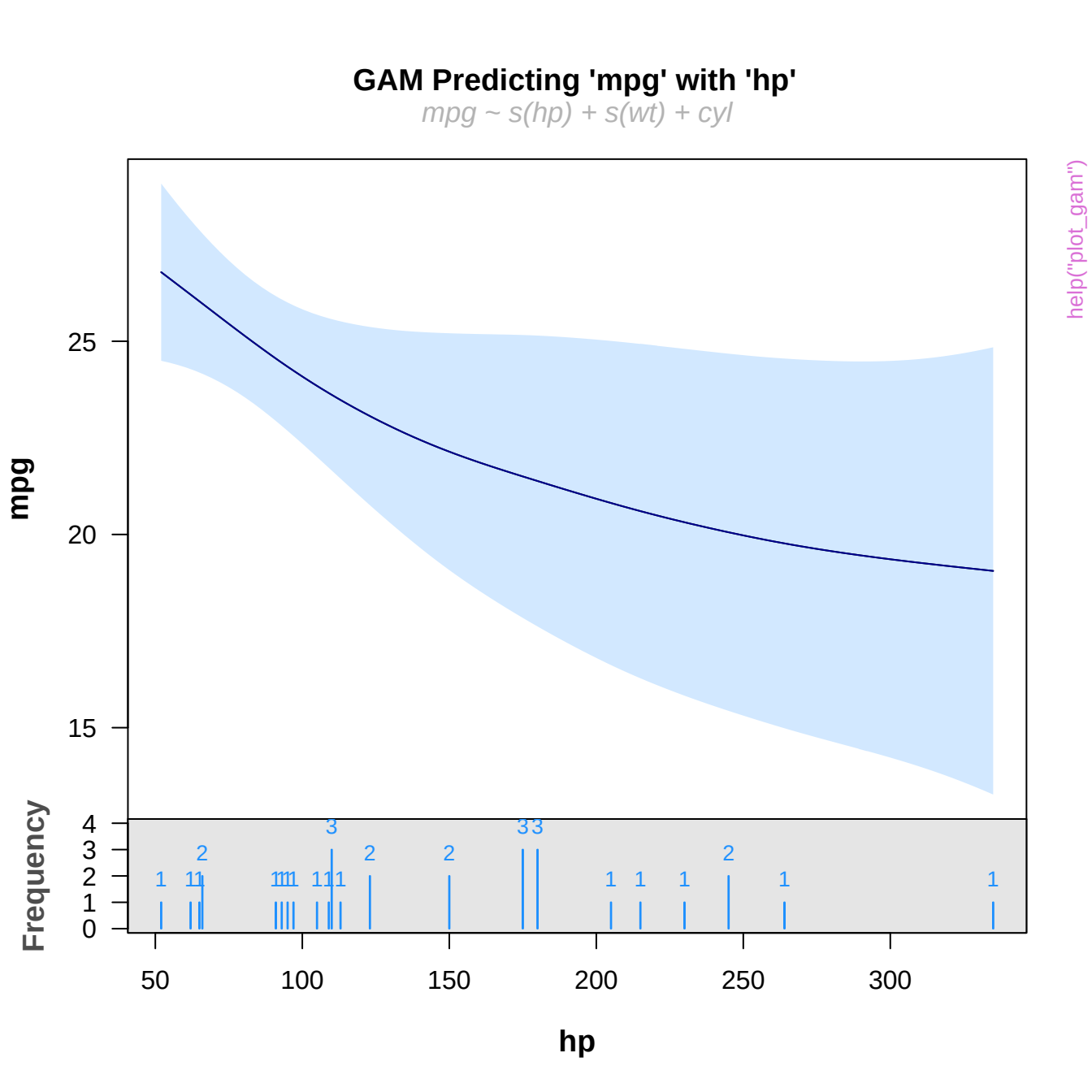

Distribution of value
(N=7)
Frequency
value
0
1
2
3
4
1
2
5
2
3
2
help("plot_freq")

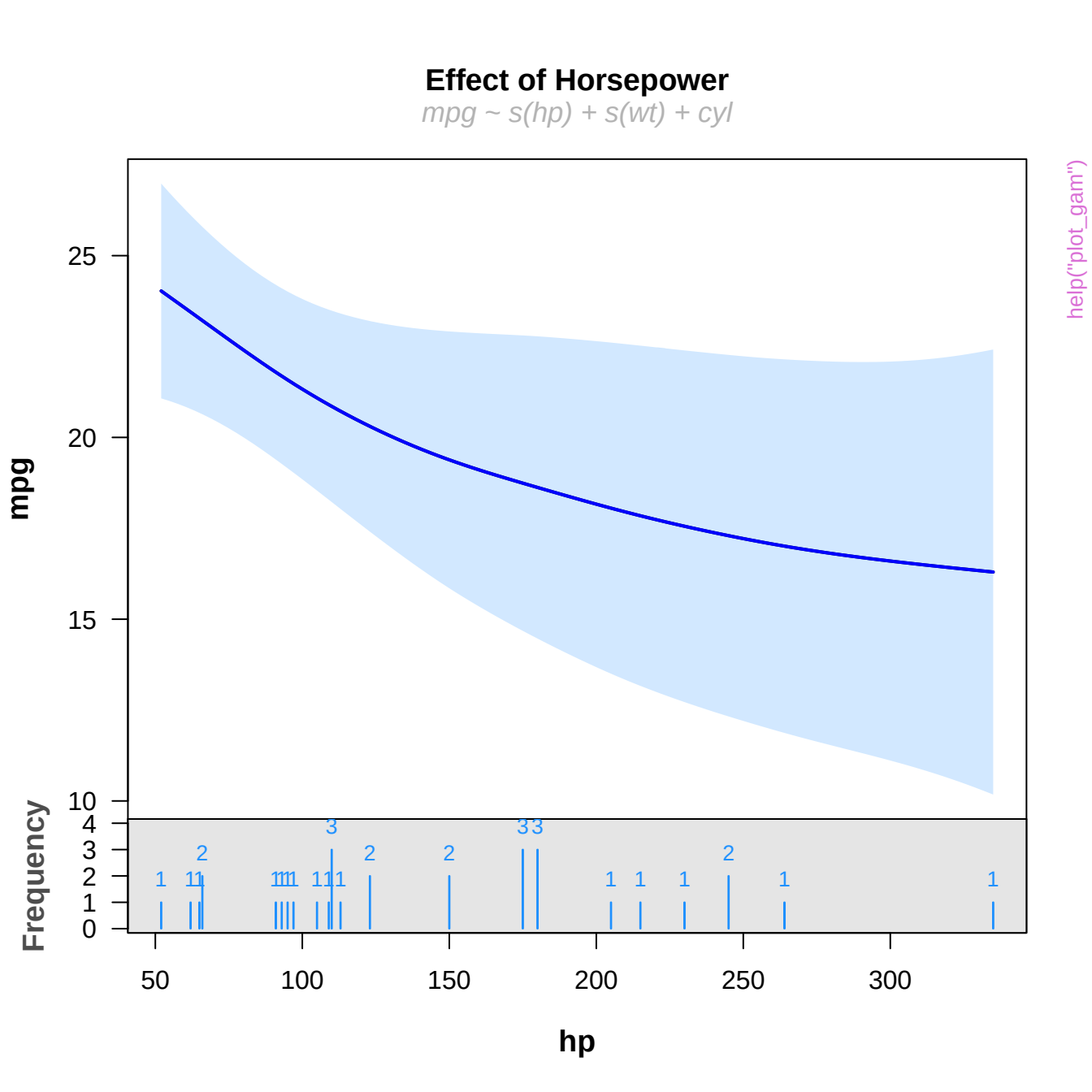

Distribution of value
(N=7)
A (N=4)
B (N=3)
help("plot_freq")
Frequency
0
1
2
2
2
1
1
1
1
2
5
value

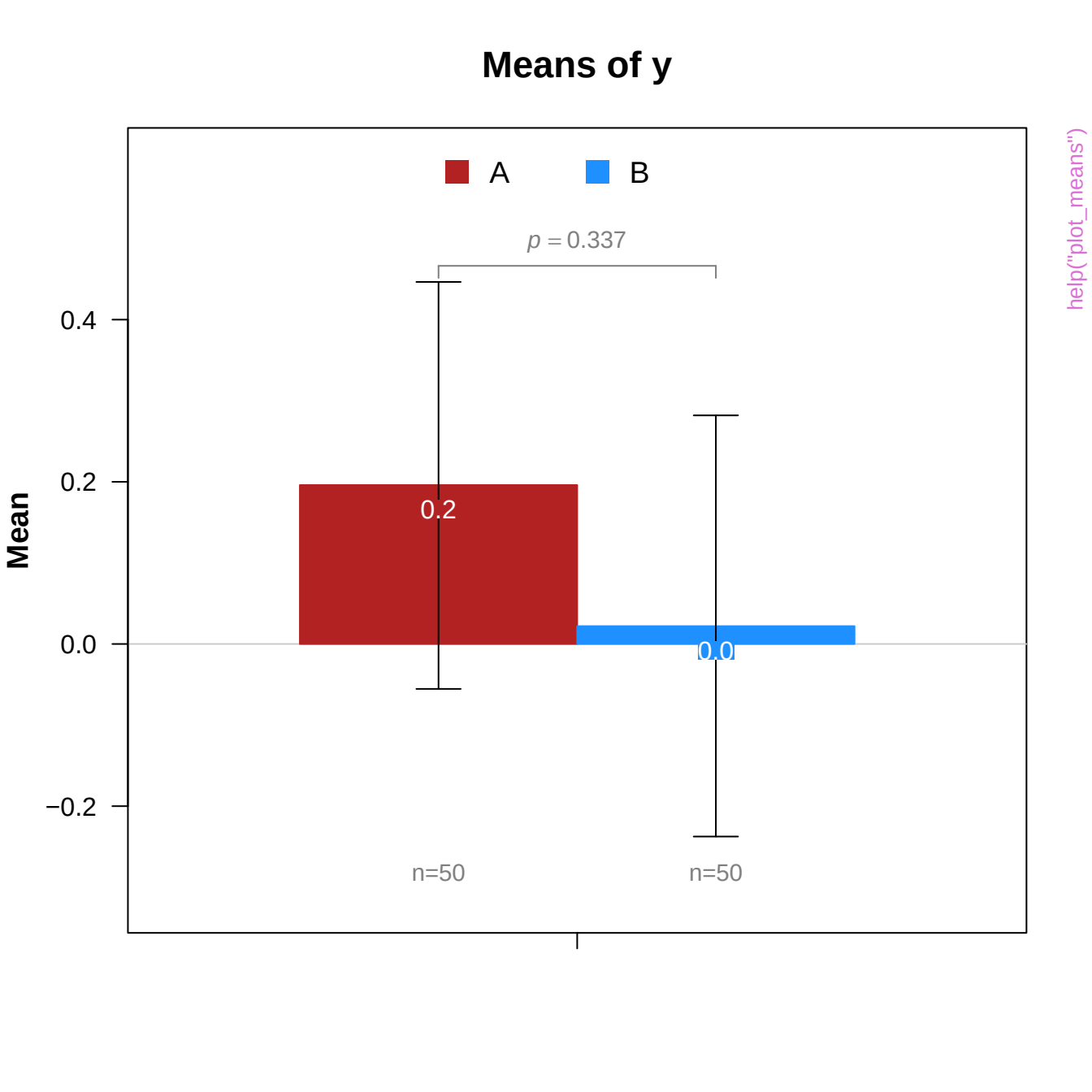

Distribution of value
(N=7)
B (N=3)
A (N=4)
help("plot_freq")
Frequency
0
1
2
2
2
1
1
1
1
2
5
value

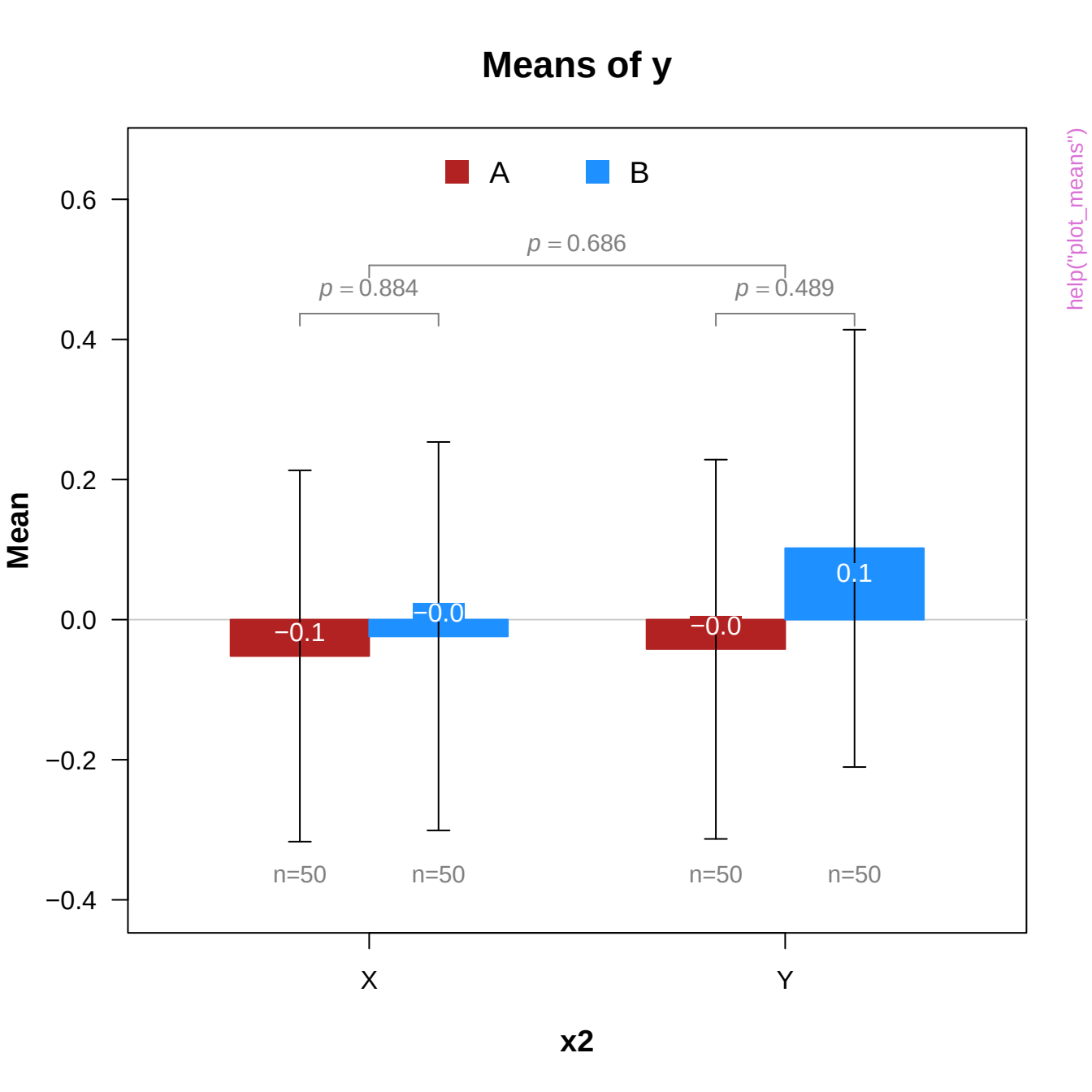

Distribution of value
(N=7)
B (N=3)
A (N=4)
help("plot_freq")
Frequency
0
1
2
2
2
1
1
1
1
2
5
value

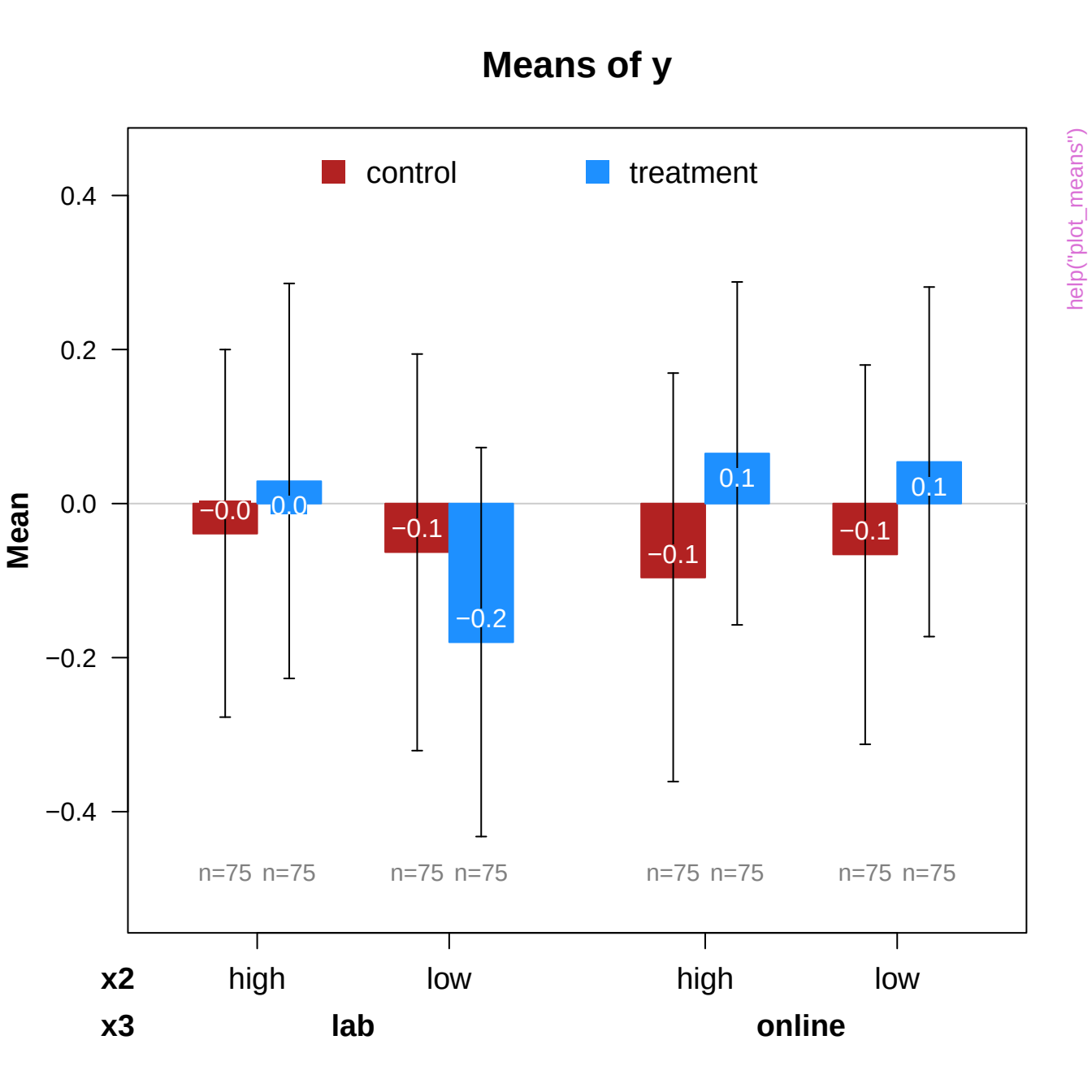

GAM Predicting 'mpg' with 'hp'
mpg ~ s(hp) + s(wt) + cyl
help("plot_gam")
mpg
25
20
15
10
Frequency
4
3
2
1
0
50
100
150
200
250
300
hp


GAM Predicting 'mpg' with 'hp'
mpg ~ s(hp) + s(wt) + cyl
help("plot_gam")
mpg
25
20
15
Frequency
4
3
2
1
0
1 1 2 1 1 1 1 1 3 1 2 2 3 3 1 1 1 2 1 1
50
100
150
200
250
300
hp


Effect of Horsepower
mpg ~ s(hp) + s(wt) + cyl
help("plot_gam")
mpg
25
20
15
10
Frequency
4
3
2
1
0
50
100
150
200
250
300
hp


Means of y
A
B
p = 0.337
0.4
0.2
0.0
−0.2
Mean
0.2
0.0
n=50
n=50
help("plot_means")


Means of y
A
B
p = 0.686
p = 0.884
p = 0.489
0.6
0.4
0.2
0.0
−0.2
−0.4
Mean
−0.1
−0.0
−0.0
0.1
n=50
n=50
n=50
n=50
X
Y
x2
help("plot_means")


Means of y
control
treatment
0.4
0.2
0.0
−0.2
−0.4
Mean
−0.0
0.0
−0.1
−0.2
−0.1
0.1
−0.1
0.1
n=75 n=75
n=75 n=75
n=75 n=75
n=75 n=75
x2
high
low
high
low
x3
lab
online
help("plot_means")